DRILL 01

PPE & SCBA

NFPA 1001 Level 1 (JPR 4.3.1, 4.3.3) / WorkSafeNB Reg 91-191

SCBA DONNING, SEAL CHECK & EMERGENCY BYPASS

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Pumper | 🦺 **PPE:** Full Structural Turnout Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs (Buddy System)

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to don and seal their SCBA within **60 seconds**, perform operational safety checks, and operate the emergency bypass valve without looking. In rural NB, initial arriving crews (often 2–3 members) cannot afford fumbling with straps or discovering an empty cylinder or leaking facepiece seal at the front door.

📦 TOOLS & APPARATUS SETUP

- Apparatus SCBA packs and facepieces (minimum 1 per member; min 4,000 PSI cylinder pressure).
- Timing device / stopwatch (for Phase 3 only).
- Disinfectant wipes and clean towels for reset.
- Set up 2–4 donning mats or staging blankets on the apparatus bay floor.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **WorkSafeNB Reg 91-191 Compliance:** Ensure no jewellery is worn. Flash hoods must be completely tucked under bunker coat collars with no skin exposed.

⚠️ **Pinch Points & Lifting:** Check for proper lifting mechanics when hoisting SCBA packs.

🛑 **THE STOP RULE: If a member shows signs of hyperventilation, dizziness, or a strap snags dangerously, shout "STOP!" to pause the evolution.**

PHASE 1: TAILGATE WALKTHROUGH (10 MINUTES) — SLOW DRY RUN

1. Drill Leader demonstrates the **Over-the-Head** and **Coat Method** donning techniques slowly.
2. Review the **Pre-Donning Check (COAT)**:
   - **C**ylinder: Full (min 4,000–4,500 PSI) and open fully.
   - **O**utlet/Regulator: Clean, HUD working, bypass valve closed (clockwise).
   - **A**ir/Harness: Straps fully extended, waist belt unbuckled.
   - **T**esting PASS: Manual alarm activation and auto-reset.
3. Review facepiece seal: Negative pressure test (cover regulator port with palm, inhale gently, hold 5 seconds; mask must collapse slightly with no inward air leaks).

DRILL 01: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

PPE & SCBA • Practical Application

PHASE 2: SLOW REPETITION & COACHING (20 MINUTES) — BUDDY COACHING

1. Firefighters pair up in buddies (Firefighter A and B).
2. Firefighter A dons gear step-by-step; Firefighter B acts as safety checker:
   - Don pants and boots.
   - Pull hood over head onto neck.
   - Don SCBA harness (tighten waist belt first to place weight on hips, then pull shoulder straps down).
   - Fasten chest strap (if equipped).
   - Don facepiece (chin in pocket, tighten lower straps then upper straps, never overtighten).
   - Conduct negative pressure seal check.
   - Pull flash hood up and over mask seal, covering all hair, ears, and neck.
   - Don helmet and fasten chinstrap snugly.
   - Don gloves. Click regulator into facepiece, take a deep breath to activate positive pressure.
   - Buddy B checks Buddy A for 100% skin coverage and strap security, then they swap.

PHASE 3: REAL-TIME CHALLENGE (15 MINUTES) — TIMED / STRESS

1. **The 60-Second Challenge:** Members start in bunker pants/boots with coat open. On the whistle, don SCBA, facepiece, hood, helmet, and gloves, and achieve positive pressure within 60 seconds.
2. **Blacked-Out Emergency Bypass Rep:** Firefighters close their eyes (or reverse their flash hood over their mask). On command, locate and open the red purge/bypass valve to deliver a continuous stream of air, then close it back to normal flow using gloved hands only.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Waist belt left unbuckled or dangling:* Shifts all pack weight onto the shoulders, causing rapid fatigue.
- ❌ *Flash hood bunched over facepiece seal:* Breaks the airtight seal and causes catastrophic air leakage.
- ❌ *Over-tightening mask straps:* Distorts the silicone seal and causes leaks around temples.
- ❌ *Forgetting to open cylinder valve completely:* Can cause low-pressure starvation during heavy exertion.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Who struggled to don gloves with wet/cold hands?", "Did anyone feel air leaking around their temple or jawline?"***

✅ APPARATUS RESET CHECKLIST

- ☐ All SCBA cylinders refilled to 4,000–4,500 PSI or swapped with full bottles from the rack.
- ☐ Regulators disinfected and clipped into their protective holders.
- ☐ Facepieces wiped dry, inspected for scratches/tears, and placed in bags.
- ☐ All harness straps (shoulders, waist, chest) fully extended/loosened for the next call.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 02

PPE & SCBA

NFPA 1001 Level 1 (JPR 4.3.1, 4.3.5) / WorkSafeNB Reg 91-191

SCBA EMERGENCY PROCEDURES & AIR MANAGEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Rescue | 🦺 **PPE:** Full Structural Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs (Buddy System)

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must instinctively handle life-threatening SCBA malfunctions (regulator freeze, runaway free-flow, out-of-air) and execute air conservation techniques (skip breathing). In rural operations with long distances from backup, mastering your air supply is the difference between walking out alive and becoming a Mayday.

📦 TOOLS & APPARATUS SETUP

- 1 SCBA per participant (with working PASS devices and cylinder pressure minimum 2,500 PSI).
- EBSS (Emergency Breathing Safety System / "Buddy Breathing") quick-connect hoses (if equipped).
- Darkened bay or hoods reversed over masks to simulate zero visibility.
- Stopwatch to time skip-breathing intervals.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Hyperventilation Warning:** When practicing skip breathing or air restriction, monitor members closely for lightheadedness or anxiety.

⚠️ **Buddy System (WorkSafeNB 51.6(2)):** Firefighters must stay within arm's reach of their partner during all low-visibility evolutions.

🛑 **THE STOP RULE: Shout "STOP!" immediately if any member experiences real respiratory distress or removes their mask in simulated zero-visibility without safety clearance.**

PHASE 1: TAILGATE WALKTHROUGH & EXPLANATION (10 MINUTES) — SLOW DRY RUN

1. **Regulator Free-Flow / Malfunction:**
   - If the regulator fails open (free-flowing air), open the red **purge/bypass valve** slightly, partially close the **main cylinder valve** to regulate flow, and breathe directly from the flow while exiting immediately.
2. **Skip Breathing / Box Breathing Technique:**
   - Inhale normally -> Hold breath for 5 seconds -> Inhale a little more -> Exhale slowly over 5 seconds.
   - Emphasize: Do not hyperventilate; slow your heart rate down.
3. **PASS Device Activation & Silencing:**
   - Demonstrate manual button press to activate full alarm.
   - Demonstrate how to hold the yellow/reset button to temporarily silence PASS when transmitting a radio Mayday.

DRILL 02: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

PPE & SCBA • Practical Application

PHASE 2: CONTROLLED PRACTICE REPETITIONS (20 MINUTES) — BUDDY COACHING

1. **Malfunction Drill (Full Gear, Lights On):**
   - Firefighter dons SCBA and breathes on air.
   - Leader simulates a stuck regulator. Firefighter:
2. Reaches back to confirm cylinder valve is fully open.
3. Cracks the bypass valve counter-clockwise.
4. Notifies partner by tapping shoulder and giving the "thumbs-out / exit" signal.
5. Transmits simulated radio report: *"Command from Interior, experiencing regulator malfunction, exiting Alpha door."*
6. **Buddy Breathing / EBSS Connection Drill:**
   - Partners kneel facing each other.
   - Firefighter A simulates zero air. Firefighter B unzips/deploys the EBSS buddy hose from their pouch.
   - Firefighter B couples into Firefighter A's intermediate pressure line until it clicks locked.
   - Both confirm steady airflow, grasp each other's harness strap with one hand, and crawl in tandem 20 metres.

PHASE 3: ZERO-VISIBILITY AIR CONSERVATION MAZE (20 MINUTES) — TIMED / STRESS

1. Reverse flash hoods over facepieces (zero visibility).
2. Partners maintain contact (hand on shoulder or 6-foot webbing strap).
3. Crew must crawl through a 30-metre obstacle course (around tires, under tables, over hose) while practicing continuous skip-breathing.
4. Time the evolution: Show firefighters how calming their breathing doubles their remaining air time compared to panic breathing.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Freezing or hesitating to call for help:* If your low-air bell/vibralert sounds, you must already be exiting, not looking for more work.
- ❌ *Forgetting to silence PASS during radio transmissions:* Screaming into the mic over a blaring PASS makes your radio traffic completely unintelligible to the pump operator or IC.
- ❌ *Separating from your partner during buddy breathing:* If you pull apart, the quick-connect can rip or disengage. Keep one hand locked onto your partner's harness.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Who felt claustrophobic or rushed their breathing?", "How easy was it to find your bypass knob with thick gloves on?"***

✅ APPARATUS RESET CHECKLIST

- ☐ All SCBA cylinders swapped and refilled to 4,000–4,500 PSI.
- ☐ PASS devices verified shut off (listen for chirps).
- ☐ Attendance roster signed.
- ☐ EBSS buddy-breathing pouches neatly folded and secured in pouches.
- ☐ Facepieces disinfected and dried.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 03

PPE & SCBA

NFPA 1001 Level 1 (JPR 4.3.1, 4.5.2) / WorkSafeNB Reg 91-191

SCBA CYLINDER REPLACEMENT (SOLO, BUDDY & CASCADE)

Time: 45–60 min | **Apparatus:** Any Engine/Rescue | **PPE:** Full Structural Gear or Work Uniform + Safety Glasses

Roles: Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs

OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to change an empty SCBA cylinder on their own pack in under 60 seconds, change their partner's cylinder while their partner remains on air with pack on (buddy swap), and safely operate the station cascade or compressor fill system. Rapid cylinder turnarounds keep working crews in service at working structure fires.

TOOLS & APPARATUS SETUP

- 4 to 6 SCBA harnesses (seated in brackets or staged on floor).
- 6 to 8 spare SCBA cylinders (empty and full).
- Station cascade fill station / compressor (or mobile cascade if equipped).
- Safety glasses / eye protection and hearing protection for fill station.

SAFETY WATCH-OUTS & STOP RULE

- **High-Pressure Hazard:** SCBA cylinders operate at 2,216 to 4,500 PSI. Never fill a cylinder without verifying hydrostatic test date and visual inspection for gouges, cracks, or burnt composite wrap.
- **Blast Door Safety:** Never fill cylinders outside the certified containment fill station blast chamber. Blast door must be locked before opening fill valves.

THE STOP RULE: Shout "STOP!" if someone attempts to fill a damaged or expired cylinder, or if high-pressure air leaks with a violent hiss.

PHASE 1: INSPECTION & SOLO CYLINDER CHANGE (15 MINUTES) — SLOW DRY RUN

1. **Inspection Points (Before Swapping):**
   - Inspect cylinder threads and the rubber high-pressure **O-ring** in the coupling. If O-ring is missing, nicked, or cracked, take cylinder out of service.
   - Verify hydrostatic test date stamped on cylinder (Carbon fiber: 5-year cycle, 15-year total life).
2. **Solo Swap Procedure (Pack on Ground):**
   - Close cylinder valve completely. Bleed residual air via regulator bypass.
   - Disconnect high-pressure coupling (hand-tighten only, never use wrenches).
   - Release cylinder retaining clamp/strap. Slide empty cylinder down and out.
   - Insert full cylinder (valve facing down toward harness outlet).
   - Lock retaining clamp. Hand-thread high-pressure coupling until snug.
   - Open cylinder valve fully, listen for air leaks, check remote gauge vs cylinder gauge (must agree within 100–200 PSI).

DRILL 03: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

PPE & SCBA • Practical Application

PHASE 2: THE TWO-PERSON "IN-SERVICE" BUDDY SWAP (20 MINUTES) — BUDDY COACHING

1. Firefighter A kneels on one knee with full gear and SCBA on their back (staying on air).
2. Firefighter B stands behind:
   - Close Firefighter A's cylinder valve.
   - Firefighter A takes 2–3 breaths to bleed pressure until alarm chimes.
   - Firefighter B unscrews the high-pressure handwheel.
   - Firefighter B unbuckles retaining strap, lifts empty cylinder out.
   - Firefighter B inspects O-ring, slides full cylinder in, snaps retaining clamp shut.
   - Firefighter B connects high-pressure coupling hand-tight.
   - Firefighter B opens cylinder valve fully, taps Firefighter A on both shoulders, and announces: *"You're full! Valve open, 4,500 PSI, good to go!"*
3. Swap roles and repeat.

PHASE 3: STATION CASCADE FILL STATION OPERATION (15 MINUTES) — TIMED / STRESS

1. Review manufacturer SOP for station cascade:
   - Inspect cylinder for damage and test date.
   - Place cylinder inside fragmentation fill tube/chamber.
   - Connect fill whip and hand-tighten bleed screw.
   - Close and latch the heavy blast chamber door.
   - Select lowest bank cylinder that has higher pressure than the tank. Open slowly (max fill rate: 300–500 PSI/min to prevent heat buildup).
   - Cascade through Banks 1, 2, and 3 until target pressure (4,500 PSI) is reached.
   - Close bank valve, close cylinder valve, open bleed valve, disconnect whip.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Cross-threading high-pressure coupling:* Ruptures threads and destroys the connection. Always align straight and hand-thread smoothly.
- ❌ *Rapid filling:* Filling too quickly overheats the cylinder. When it cools down in the truck, pressure drops 500+ PSI, leaving you short on air!
- ❌ *Dropping cylinders on concrete:* Carbon-wrapped bottles can suffer invisible structural delamination. Never let cylinders tip over.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How easy was it to guide the cylinder into the harness bracket while your partner was kneeling?", "Did we find any worn O-rings?"***

✅ APPARATUS RESET CHECKLIST

- ☐ All empty training cylinders refilled and placed back into apparatus racks.
- ☐ Cascade compressor turned off, bank valves isolated.
- ☐ Eye protection returned to storage.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 04

GROUND LADDERS

NFPA 1001 Level 1 (JPR 4.3.6, 4.3.12) / WorkSafeNB Reg 91-191

GROUND LADDERS – CARRIES, RAISES & PLACEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Ladder Rack | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | 2-Firefighter Teams

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to remove a 24-ft or 28-ft extension ladder from the apparatus, carry it safely using 1-person and 2-person methods, raise and extend it to a target window or roofline, lock the dogs, tie off the halyard, and set the proper 75-degree climbing angle. Ground ladders are the primary secondary egress for interior crews operating in two-story rural dwellings.

📦 TOOLS & APPARATUS SETUP

- Apparatus ground ladder complement: 24-ft or 28-ft two-section extension ladder, 14-ft roof ladder.
- Station exterior wall, drill tower, or suitable clear wall structure without overhead wires.
- Ladder staging area clear of potholes, gravel, or ice.

⚠️ SAFETY WATCH-OUTS & STOP RULE

- ⚠️ **Overhead Electrical Wires:** Always look up **360 degrees** before removing or raising any ground ladder. Maintain minimum 3-metre (10-foot) clearance from power lines.
- ⚠️ **Pinch Points & Dog Locks:** Keep fingers wrapped around ladder beams—never grasp rungs while raising or lowering the fly section.
- ⚠️ **Footing & Heel Placement:** The "heeler" must place their steel-toed boots against the ladder spurs and pull back on the beams while the extension is raised.

🛑 **THE STOP RULE:** Shout **"STOP!"** immediately if the ladder begins to twist, if footing slips, or if anyone touches an overhead hazard.

PHASE 1: REMOVAL, INSPECTION & CARRIES (15 MINUTES) — SLOW DRY RUN

1. **Apparatus Removal:** Release ladder retaining bracket. 2 firefighters lift ladder and slide it clear of apparatus bed.
2. **Two-Firefighter Low-Shoulder Carry:**
   - Firefighter at the butt (heel) faces direction of travel. Firefighter at the tip faces direction of travel.
   - Both place upper arm between rungs, resting the beam on the shoulder.
   - Butt firefighter calls: *"Ready to lift... Lift! Ready to walk... Forward!"*
3. **One-Firefighter Low-Shoulder Carry (14-ft roof or 24-ft flat carry):**
   - Center on ladder balance point, reach through rungs, lift onto shoulder, stabilize tip with leading hand.

DRILL 04: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ground Ladders • Practical Application

PHASE 2: THE TWO-FIREFIGHTER EXTENSION LADDER RAISE (20 MINUTES) — BUDDY COACHING

1. **Positioning:**
   - Butt firefighter places ladder heel on ground at proper distance from wall (~6 feet for a 24' ladder raised to second story).
   - Butt firefighter stands on the bottom rung or places both boots against the spurs, grasping beams firmly at chest level.
2. **Beam Raise (or Flat Raise):**
   - Tip firefighter lifts ladder above head and walks hand-over-hand along the rungs/beams toward the butt, pushing ladder to vertical.
3. **Extension & Dog Lock:**
   - Butt firefighter stabilizes vertical ladder, eyes looking up.
   - Tip firefighter steps to halyard side, grasps rope hand-over-hand, and pulls smoothly to extend fly section.
   - Ensure fly is oriented according to manufacturer specification (usually **fly out** for modern aluminum extension ladders).
   - Once tip reaches 3–5 rungs above roofline (or sill level for rescue), lower slightly until **ladder dogs (pawls) click and lock onto rungs**.
4. **Halyard Tie-Off:**
   - Wrap halyard around the rung below the dogs and tie off with a **clove hitch** and half hitch around the beam.
5. **Setting Angle (1:4 Ratio / 75 Degrees):**
   - Butt firefighter stands straight with toes touching ladder spurs, reaches arms straight forward. Palms should comfortably rest on the ladder rung at shoulder height.

PHASE 3: LADDER PLACEMENT SCENARIOS (15 MINUTES) — TIMED / STRESS

Rotate crews through 3 tactical placement positions:

1. **Window Rescue:** Tip placed directly at or just below the windowsill for rapid victim removal.
2. **Ventilation / Entry:** Tip placed upwind side of window, level with upper third of window frame.
3. **Roof Access:** Tip extended minimum **3 to 5 rungs above roof eaves** to provide visible handhold for roof crews in dark conditions.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Looking down while extending:* The heeler must keep their eyes on the tip to watch for wires and ensure dogs lock properly.
- ❌ *Fingers on rungs during lowering:* If halyard slips, fly drops instantly—shearing off fingers! Keep hands strictly on the beams.
- ❌ *Shallow climbing angle (<70°):* Causes ladder base to kick out under load. Always verify the arm-reach angle check.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did anyone forget the halyard knot?", "Was the climbing angle verified before stepping onto the bottom rung?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Fly section lowered smoothly; dogs inspected for spring tension.
- ☐ Halyard neatly coiled and tied to bottom rungs.
- ☐ Ladder beams wiped clean of grit and road salt.
- ☐ Ladder stowed into apparatus rack and locked with safety pins.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 05

GROUND LADDERS

NFPA 1001 Level 1 (JPR 4.3.6, 4.3.12) / WorkSafeNB Reg 91-191

ROOF LADDER DEPLOYMENT & LEG LOCK WORK POSITION

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Ladder Rack | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Ladder Climber & Ground Heeler

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to carry a 14-ft or 16-ft roof ladder up an extension ladder, open and lock the spring hooks, slide the ladder up a pitched roof until the hooks catch the roof ridge, and transition safely onto the roof ladder while executing a secure **leg lock** or life-safety harness tie-in. Essential for rural chimney fires and roof ventilation on steep Maritime roofs.

📦 TOOLS & APPARATUS SETUP

- 24-ft or 28-ft extension ladder raised and secured to station roofline or training prop (minimum 3–5 rungs above gutter).
- 14-ft or 16-ft roof ladder with spring-loaded folding hooks.
- Flathead axe or halligan tool for sounding roof deck.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Fall Arrest / Ladder Safety:** The base extension ladder must be firmly heeled or tied off at the base. Climbers must maintain **3 points of contact** at all times.

⚠️ **Hook Lock Verification:** Visually and physically confirm both roof hooks have clicked and locked open before the ladder leaves the ground.

⚠️ **Ridge Embedment:** The firefighter must push the roof ladder over the ridge and pull back firmly until both hooks bite deeply into the roof ridge board.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the climber loses balance, if the roof ladder fails to catch the ridge, or if footing on the roof deck slips.**

PHASE 1: GROUND PREPARATION & HOOK DEPLOYMENT (10 MINUTES) — SLOW DRY RUN

1. Remove roof ladder from apparatus.
2. Inspect spring-loaded hooks: Pull outwards and pivot 180 degrees until the locking pins snap into place.
3. Test hook locking mechanism by hand pressure; confirm hooks do not collapse.

DRILL 05: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ground Ladders • Practical Application

PHASE 2: CARRYING ROOF LADDER UP GROUND LADDER (20 MINUTES) — BUDDY COACHING

1. **The Shoulder Carry Method:**
   - Firefighter faces the extension ladder with roof ladder resting on their shoulder, hooks pointing forward and curled outward/downward.
   - Grasp the rungs of the extension ladder with both hands while sliding the roof ladder along on the shoulder.
   - Climber climbs smoothly, taking one rung at a time.
2. **Transition at the Eaves:**
   - Stop when your shoulders are level with the roof eaves.
   - Rest the butt of the roof ladder onto the roof deck or gutter edge.
   - Push the roof ladder upward with both hands along the roof surface, allowing the wheels (if equipped) or hooks to slide smoothly up to the peak.
   - As the hooks clear the ridge peak, rotate or pull the ladder downward firmly until both hooks **bite into the opposite side of the roof ridge**.
   - Pull hard 2–3 times to verify total engagement before placing any body weight on it.

PHASE 3: TRANSITION & LEG LOCK WORK POSITION (20 MINUTES) — TIMED / STRESS

1. **Transitioning onto Roof Ladder:**
   - Step carefully from the extension ladder onto the center rungs of the roof ladder (never step directly onto un-sounded roof deck).
   - Keep your weight centered over the beams.
2. **Executing the Leg Lock (Hands-Free Work Position):**
   - Reach your leg through the ladder rungs (one rung above your standing foot).
   - Wrap your lower leg around the beam and hook your foot behind the rung below.
   - Straighten your other leg to lock your shin and instep securely against the ladder.
   - Step back with your hands to demonstrate a secure, hands-free position (simulating operating an axe, TIC, or chimney drop-chain).
3. If using ladder safety belts, connect the carabiner/hook to the center of the rung at waist height.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Hooks not locked before ascending:* Trying to deploy hooks with gloved hands while dangling from a 20-foot ladder is dangerous and slow. Lock them on the ground!
- ❌ *Stepping between rungs on the roof deck:* Rotted or fire-weakened roof sheathing will collapse under your boots. Always step strictly on the ladder rungs.
- ❌ *Pushing ladder without looking at the ridge:* If you miss the ridge or hit a chimney flashing, the ladder can slide down and kick you off the extension ladder.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Who felt unstable while transferring from the extension ladder to the roof ladder?", "Did the hooks bite solid into the peak?"***

✅ APPARATUS RESET CHECKLIST

- [ ] Roof ladder hooks released and folded flat against beams.
- [ ] Spring-loaded pins lubricated if stiff.
- [ ] Extension ladder lowered and both ladders returned to brackets.
- [ ] Safety straps latched.
- [ ] Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 06

HOSE & STREAMS

NFPA 1001 Level 1 (JPR 4.3.7, 4.3.10) / WorkSafeNB Reg 91-191

ATTACK LINE DEPLOYMENT & FLAKING (CROSSLAY / MINUTEMAN)

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Nozzleman & Backup Firefighter

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to rapidly deploy a 200-ft (60m) 1.75" (45mm) preconnected attack line from the apparatus crosslay or rear bed, stretch it to the entrance doorway, flake out the 50-ft working length without kinks, call for water, and bleed air from the nozzle before entering. A clean stretch guarantees maximum water flow without pressure-robbing spaghetti kinks.

📦 TOOLS & APPARATUS SETUP

- Apparatus equipped with 1.75" (45mm) preconnects (Minuteman, Crosslay, or Triple-Layer load).
- Attack nozzle (combination fog or smooth bore).
- Cones or simulated structure entrance doorway 100–150 feet away from the apparatus.
- Pump operator or member staged at pump panel to charge the line.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Tripping Hazards:** Watch for firefighters tripping backward over hose folds during the stretch.

⚠️ **Kink Check:** When line charges, inspect full length for sharp 90-degree kinks (each severe kink cuts water flow by up to 30%).

⚠️ **Nozzle Reaction:** Ensure the nozzle firefighter has a firm two-handed grip with bail closed before calling for water.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the hose violently kinks, whips, or knocks a member off balance while charging.**

PHASE 1: WALKTHROUGH OF LOAD & DEPLOYMENT MECHANICS (15 MINUTES) — SLOW DRY RUN

1. **Understand the Load (Minuteman / Crosslay):**
   - Review how your department's hose bed is packed.
   - Identify the **nozzle loop** (working length) and the **pull loop(s)** (drag load).
2. **The 3 Rules of Hose Deployment:**
   - **Rule 1:** Take your time on the pull; speed comes from clean deployment, not running.
   - **Rule 2:** The nozzle and working length (last 50 ft) MUST reach the threshold of the entry door before the line is charged.
   - **Rule 3:** Flake the working length in large, loose loops parallel to the building or back down the driveway—never dump a pile of hose directly on the front porch.

DRILL 06: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Hose & Streams • Practical Application

PHASE 2: DEPLOYMENT & FLAKING REPETITIONS (20 MINUTES) — BUDDY COACHING

1. **Execution Steps:**
   - **Step 1 (The Pull):** Step up to the crosslay/rear bed. Grasp the nozzle and nozzle loop with your dominant hand. Grasp the second pull loop with your off-hand.
   - **Step 2 (The Stretch):** Pull all hose clear of the bed. Place the nozzle load onto your shoulder. Turn and walk toward the designated entry door.
   - **Step 3 (Dropping the Drag):** When the drag load runs tight (~75–100 ft out), drop the secondary loops from your hand. Continue walking with the shoulder load to the door.
   - **Step 4 (Flaking the Working Length):** Stop 10–15 feet from the door. Lay the nozzle down gently. Walk the remaining loops toward the open driveway or yard, flaking the hose in an "S" or "U" pattern free of twists.
2. **Calling for Water & Bleeding the Line:**
   - Nozzleman picks up nozzle, sits/anchors the line behind their hip, and calls via radio: *"Engine 1, charge the 1.75 crosslay!"*
   - As water arrives, watch the line stiffen. Shake out any minor kinks.
   - Point nozzle upward at a 45-degree angle. Open the bail slowly to bleed trapped air until a solid, unbroken stream flows.
   - Check nozzle pattern (straight stream vs narrow fog) and close bail smoothly.

PHASE 3: TIMED CREW STRETCH (15 MINUTES) — TIMED / STRESS

1. Team of 2 (Nozzleman & Backup Firefighter) start seated in the cab or at the tailboard.
2. On call: *"Deploy attack line to Side Alpha front door!"*
3. Deploy, stretch, flake, charge, bleed air, and announce *"Line charged and ready for entry!"* Target time: **under 90 seconds**.
4. Reverse roles and repeat with alternate hose loads.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Charging the line in the hose bed:* Never call for water while hose is still packed in the tray—it expands and locks itself into the bed!
- ❌ *Dropping the nozzle short of the door:* If the nozzle drops 50 feet from the door, you must drag 200 lbs of water-filled hose just to reach the threshold.
- ❌ *Failing to bleed air before entering:* Trapped air causes violent nozzle sputtering and momentary water loss during an interior attack.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the working length flake cleanly without twisting?", "How many kinks did the safety officer find in the lay?"***

✅ APPARATUS RESET CHECKLIST

- [ ] Line drained: Disconnect nozzle, open bleeder, walk hose from fold to coupling to push out all water.
- [ ] Couplings inspected for dirt/grit in threads and intact rubber gaskets.
- [ ] Hose repacked carefully to department standard (Minuteman / Crosslay) with pull loops properly spaced.
- [ ] Nozzle attached, bail shut, gallonage set.
- [ ] Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 07

HOSE & STREAMS

NFPA 1001 Level 1 (JPR 4.3.10) / WorkSafeNB Reg 91-191

HOSE HANDLING, BODY MECHANICS & ADVANCEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Nozzleman & Backup Firefighter

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must master modern hose handling mechanics (the **Hip Grip** or **Arm-Clamp** technique) to absorb 60–80 lbs of nozzle reaction force without fatigue, coordinate advancement with a backup firefighter, negotiate 90-degree corners and stairs, and maintain strict door control. Proper technique prevents fatigue and prevents nozzlemen from being thrown off balance.

📦 TOOLS & APPARATUS SETUP

- 1 charged 1.75" (45mm) attack line (100–150 ft) flowing at normal fireground operating pressure (~100–125 PSI discharge pressure, delivering 150–160 GPM / 600 LPM).
- Simulated corner/hallway (between apparatus bays, around building corners, or training door prop).
- Webbing strap (for door control).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Back Strain & Ergonomics:** Watch for nozzlemen leaning backward or fighting nozzle reaction with bent arms. The nozzle reaction must be driven straight down into the hips and legs.

⚠️ **Pinch Points & Knuckles:** Ensure hands are not dragged along brick/door frames when navigating tight corners under pressure.

⚠️ **Water Runoff:** Direct runoff toward exterior storm drains or open gravel; monitor for slick surfaces in freezing weather.

🛑 **THE STOP RULE: Shout "STOP!" if the nozzle firefighter loses grip or control of the bail, or if a member falls.**

PHASE 1: MODERN NOZZLE MECHANICS (THE HIP GRIP) (15 MINUTES) — SLOW DRY RUN

1. **The Flaw of the "Death Grip":**
   - Grasping the nozzle with two hands held out in front of the chest burns out shoulder and arm muscles in under 2 minutes.
2. **The Modern Hip Grip Technique:**
   - Tuck the charged hose line firmly between your hip and upper arm (or lock it under your armpit against your ribcage).
   - Rest your forward hand on the hose 6–8 inches behind the nozzle coupling.
   - Rest your rear hand on top of the nozzle shutoff bail.
   - Lean forward aggressively with your front knee bent and back leg braced (athletic boxer stance).
   - The backward thrust of the water stream is absorbed directly by your core and hips into the ground.

DRILL 07: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Hose & Streams • Practical Application

PHASE 2: THE BACKUP FIREFIGHTER'S ROLE (15 MINUTES) — BUDDY COACHING

1. **Positioning:**
   - The backup firefighter positions on the **same side of the hose** as the nozzleman, 3 to 4 feet behind.
   - Lock your forward arm around the hose and rest your back hand firmly against the nozzleman's SCBA cylinder harness or shoulder.
2. **Absorbing Reaction & Pushing Forward:**
   - The backup absorbs any remaining backward reaction, allowing the nozzleman to direct the stream freely with minimal effort.
   - When moving forward, the backup firefighter provides the forward drive: *"Moving forward in 3, 2, 1... Push!"*

PHASE 3: CORNER NAVIGATION & DOOR CONTROL ADVANCEMENT (20 MINUTES) — TIMED / STRESS

1. **Navigating a 90-Degree Corner:**
   - Never let the hose pin the nozzleman against the outside wall of a corner.
   - The backup firefighter drops to one knee at the corner vertex, becoming a **"human pulley"**, holding the hose off the wall so slack feeds freely to the nozzleman.
   - Nozzleman advances 10–15 feet into the room, sweeps the ceiling and floor, and calls for more line.
2. **Door Control & Chocking:**
   - Before entering, firefighter checks the door for heat with gloved hand/TIC.
   - Chock the door open using a wooden wedge or hose strap so the door cannot pinch the charged hose and cut off water!

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Standing upright while flowing:* Standing tall raises your center of gravity; nozzle reaction will push you backwards. Stay low and lean into the stream.
- ❌ *Backup firefighter on opposite side:* Straddling the hose or standing on opposite sides creates a scissors hazard if the line whips or snags.
- ❌ *Letting the door close on the hose:* A self-closing spring hinge will pinch a charged 1.75" line against the jamb, choking water flow to a trickle. Always chock the door!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How big of a difference did the hip grip make in muscle fatigue?", "How well did the backup firefighter communicate forward movement?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Nozzle closed smoothly to prevent water hammer.
- ☐ Pump throttle brought to idle; discharge gate closed.
- ☐ Hose drained, rolled, and repacked.
- ☐ Door wedges returned to gear pockets.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 08

HOSE & STREAMS

NFPA 1001 Level 1 (JPR 4.3.10) / WorkSafeNB Reg 91-191

EXTENDING A WORKING HOSELINE (BREAK & EXTEND)

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to extend an active, charged attack line that has come up short of the fire room without requiring the pump operator to shut down the entire discharge at the truck. Using the **Break-and-Extend** method or a manual **Hose Clamp / Z-Kink**, crews can add a 50-ft section in under 45 seconds right at the doorway.

📦 TOOLS & APPARATUS SETUP

- 1 charged 1.75" (45mm) attack line (100–150 ft) with combination or breakaway nozzle (shutoff ball valve with threaded 1.5" NST outlet behind tip).
- 1 spare 50-ft (15m) section of 1.75" hose rolled or flaked.
- 1 manual hose clamp (or training technique using double/Z-kink).
- Spare nozzle or original nozzle tip.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **High-Pressure Release:** When uncoupling a clamped or kinked hose, residual pressure can spray high-velocity water into faces. Ensure eye protection/visors are down.

⚠️ **Hose Clamp Pinch Points:** Keep fingers completely clear of mechanical clamp jaws and leverage levers.

🛑 **THE STOP RULE: Shout "STOP!" immediately if a kinked line slips loose or if high-pressure water whips uncoupled hose.**

PHASE 1: UNDERSTANDING THE BREAKAWAY NOZZLE & THE SHORT STRETCH (10 MINUTES) — SLOW DRY RUN

1. In rural residential structures (large farmhouses, split-levels, long driveways), 200 feet of preconnect sometimes falls 20–30 feet short of the second-story bedroom or attic.
2. If your department runs **breakaway nozzles**, the fog or smooth bore tip unscrews from the ball-shutoff valve, exposing a standard 1.5" male hose thread!
3. If not using a breakaway nozzle, the line must be clamped or z-kinked 10–15 feet behind the nozzle coupling.

DRILL 08: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Hose & Streams • Practical Application

PHASE 2: METHOD 1 – BREAKAWAY NOZZLE EXTENSION (15 MINUTES) — BUDDY COACHING

1. Nozzleman reaches maximum stretch: *"Line is short! Extending 50 feet!"*
2. Close the nozzle bail completely.
3. Unscrew the outer tip from the shutoff valve. The shutoff valve now acts as a closed gate valve holding back full line pressure.
4. Firefighter B connects the female coupling of a fresh 50-ft section directly onto the shutoff valve threads.
5. Stretch and flake the new 50-ft section into the structure.
6. Connect the nozzle tip (or a second nozzle) to the end of the new section.
7. Open the original shutoff valve slowly. Trapped air bleeds through the new nozzle; the line is now 50 feet longer and back in service!

PHASE 3: METHOD 2 – THE Z-KINK / HOSE CLAMP METHOD (20 MINUTES) — TIMED / STRESS

(Used when the nozzle is a one-piece unit without breakaway threads)

1. **The Z-Kink (Manual Clamp by Crew):**
   - Nozzleman and backup firefighter step 10 feet back from the nozzle.
   - Fold the hose back onto itself twice, creating a tight "Z" shape.
   - Both firefighters kneel onto the folds, putting their full body weight onto the kinks. The sharp folds choke off water pressure completely.
2. **Uncoupling & Connecting:**
   - A third firefighter (or nozzleman reaching forward) opens the nozzle bail to bleed pressure, then unscrews the nozzle from the coupling.
   - Attach the new 50-ft roll to the open coupling.
   - Attach the nozzle to the end of the new section.
   - On command: *"Releasing kink in 3, 2, 1!"* Firefighters slowly unbend the Z-kink, restoring full line pressure smoothly.
3. Practice with a mechanical **Hose Clamp** applied 15–20 feet back from the nozzle (always apply clamp with handle pointed toward the pump).

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Trying to unscrew a pressurized coupling:* Without bleeding line pressure through the nozzle bail first, the threads are locked by 100+ PSI of hydraulic force and will not turn.
- ❌ *Releasing clamp or kink too quickly:* Slamming water back into an empty 50-ft section creates a violent pressure spike and water hammer. Open the clamp or release the kink slowly!
- ❌ *Dropping the uncoupled nozzle in mud:* Dirt inside the nozzle tip clogs internal stream shapers and ruins fog patterns.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the Z-kink hold back pressure without slipping?", "How much faster is a breakaway nozzle extension compared to shutting down the pump panel?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Nozzle tips reattached and torqued hand-tight.
- ☐ Spare 50-ft section drained, rolled in donut or straight roll, and returned to storage.
- ☐ Hose clamp returned to apparatus bracket.
- ☐ Gaskets checked in all broken couplings.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 09

WATER SUPPLY

NFPA 1001 Level 1 (JPR 4.3.14, 4.3.15) / WorkSafeNB Reg 91-191

LARGE DIAMETER HOSE (LDH) & HYDRANT / SUPPLY LAYS

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Hydrant Firefighter & Driver/Apparatus Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to execute a **Forward Supply Lay** using 4-inch or 5-inch (100mm/125mm) Large Diameter Hose (LDH) with sexless Storz couplings, connect to a pressurized municipal hydrant or dry hydrant, flush the hydrant, make the connection with a hydrant gate valve, and charge the supply line without water hammer.

📦 TOOLS & APPARATUS SETUP

- Pumper with LDH supply bed (minimum 200–300 ft loaded).
- Hydrant kit/bag: Hydrant wrench, 2.5" gate valve, 4"/5" Storz-to-steamer adapter, rubber mallet, spanner wrenches.
- Station training hydrant or simulated hydrant marker.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Snag & Flail Hazard:** When laying LDH from a moving apparatus, ensure the hydrant firefighter stands firmly in the safety zone (behind the hydrant or off the road curb) and never steps into the loop of paying-out hose.

⚠️ **Water Hammer:** Never open or close a hydrant or master intake gate rapidly. Minimum 30 seconds to open; minimum 30 seconds to close!

⚠️ **Storz Locking Latch:** Ensure the locking lever on the Storz coupling snaps shut. If not locked, the coupling can twist and decouple violently under 150 PSI!

🛑 **THE STOP RULE: Shout "STOP!" immediately if the paying-out hose catches on a firefighter's boot or gear.**

PHASE 1: THE HYDRANT BAG & STORZ COUPLING MECHANICS (10 MINUTES) — SLOW DRY RUN

1. **The Storz Advantage:** Sexless couplings (quarter-turn twist). Align the locking lugs, push firmly together, and rotate 1/3 turn until the spring-loaded lock clicks.
2. **The Hydrant Kit Inventory:**
   - Hydrant wrench (pentagonal socket).
   - 4" or 5" Storz-to-thread adapter.
   - 2.5" gate valve (allows a second line to be connected later without shutting the hydrant down).

DRILL 09: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Water Supply • Practical Application

PHASE 2: THE FORWARD LAY (CATCHING THE HYDRANT) (20 MINUTES) — BUDDY COACHING

1. **Dismounting & Wrapping the Hydrant:**
   - Apparatus stops 15–20 feet past the hydrant.
   - Hydrant firefighter steps off apparatus with hydrant bag, grasps LDH coupling, and pulls 10–15 feet of hose off the tailboard.
   - Wrap the hose completely around the barrel of the hydrant in a secure loop.
   - Stand with a boot firmly planted on the crossed hose to anchor it against the ground.
   - Signal the driver with a loud voice and raised arm: *"LAY THE LINE! GO GO GO!"*
   - As the apparatus drives forward at 8–10 km/h, the hose pays out of the bed smoothly.
2. **Hydrant Preparation & Connection:**
   - Unwrap the hose from the hydrant barrel.
   - Remove the steamer port cap using the hydrant wrench.
   - **Flush the Hydrant:** Crack the operating nut on top 2–3 turns until water flows clear of gravel and rust, then close.
   - Thread the 4"/5" Storz adapter onto the steamer port and tighten with spanner.
   - Thread a 2.5" gate valve onto one of the side ports (closed position).
   - Connect the LDH Storz coupling to the adapter; verify locking latch snaps shut.

PHASE 3: CHARGING THE SUPPLY LINE & AIR BLEEDING (15 MINUTES) — TIMED / STRESS

1. Await radio confirmation from the pump operator: *"Engine 1 is ready for hydrant water. Charge the supply line!"*
2. Rotate the hydrant top operating nut counter-clockwise slowly until fully open (feel for the valve to hit its stop, then back off 1/4 turn to prevent sticking).
3. Pump operator opens the pump intake bleeder valve to exhaust air before opening the master intake valve.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Stepping inside the hose loop:* If the hose snags on the tailgate while the truck drives away, an internal loop will whip around your legs and cause severe fractures. Stay outside the loop!
- ❌ *Not flushing the hydrant:* Rocks, sediment, and rust sitting inside the hydrant barrel will shoot into the fire pump, destroying the impeller and packing seals.
- ❌ *Closing the hydrant fast:* Slamming a hydrant shut creates a catastrophic pressure wave (water hammer) that can rupture municipal mains under the street!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the Storz lock snap closed properly?", "How long did it take between dismounting the truck and having the hydrant connected?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Hydrant shut completely until operating nut stops; check barrel drainage (suction check).
- ☐ Caps replaced and tightened hand-tight with wrench.
- ☐ LDH drained: Walk the hose with a drainage roller or over-the-shoulder walk.
- ☐ LDH repacked in the apparatus bed (flat load with folds slightly staggered).
- ☐ Hydrant kit repacked and returned to apparatus compartment.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 10

RURAL WATER SUPPLY

NFPA 1001 Level 1 (JPR 4.3.15) / FUS STSS Standard

RURAL DROP TANK DEPLOYMENT & TANKER DUMP OPERATIONS

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Pumper & Tanker (Mobile Water Supply) | 🦺 **PPE:** Full Gear or Work Gloves/Boots

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Drop-Site Crew & Tanker Driver

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to remove a 1,500 to 3,000-gallon portable folding drop tank (Porta-Tank) from apparatus racks, deploy it on level ground, stage a low-level strainer, set up a jet siphon between dual tanks, and guide a tanker into position to dump 1,500+ gallons in under 90 seconds. Mastering this drill is mandatory to pass and maintain **Fire Underwriters Survey (FUS) Superior Tanker Shuttle Service (STSS)** accreditation in New Brunswick.

📦 TOOLS & APPARATUS SETUP

- Pumper with hard suction hose (6-inch / 150mm) and low-level strainer with air-bleed float.
- Tanker with portable folding drop tank (2,000–3,000 gal) and 10-inch rear or swivel side dump valve.
- Jet siphon attachment and 1.75" discharge line.
- Ground tarp to protect drop tank liner from sharp gravel/debris.

⚠ SAFETY WATCH-OUTS & STOP RULE

⚠ **Backing Apparatus Hazard:** The tanker must have a designated backing guide wearing a high-visibility vest. The guide must stay in the driver's left-side mirror at all times!

⚠ **Pinch Points:** Keep hands clear of folding tank scissor frames when opening. Frame hinges can pinch or crush fingers.

⚠ **Water Weight:** A 2,500-gallon tank weighs over 25,000 lbs (11,300 kg) when full! Once filled, it cannot be pushed or moved without draining.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the backing tanker loses sight of the guide or if the tank begins to collapse due to uneven ground slope.**

PHASE 1: SITE SELECTION & TANK DEPLOYMENT (15 MINUTES) — SLOW DRY RUN

1. **Site Selection:**
   - Choose flat, firm ground adjacent to the attack pumper.
   - Ensure adequate lane clearance (minimum 12 feet) for incoming tankers to pull past, dump, and exit without turning around.
2. **Deploying the Folding Tank:**
   - Firefighters lay a heavy protective canvas/vinyl ground tarp on the road surface to prevent gravel punctures.
   - Release retaining straps and lift the folding tank off the tanker rack (2–4 firefighters).
   - Stand the folded tank vertically on its bottom rails.
   - Two firefighters grasp opposite corners of the frame, pull outward smoothly until frame locks open in a square/rectangle.
   - Push the internal vinyl/hypalon liner down into the corners; ensure the drain sleeve is cinched tight and tied to the top rail!

DRILL 10: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Rural Water Supply • Practical Application

PHASE 2: LOW-LEVEL STRAINER & HARD SUCTION STAGING (15 MINUTES) — BUDDY COACHING

1. Connect low-level strainer to 10-foot section of 6-inch hard suction hose.
2. Lower strainer into the drop tank until it sits flat on the tank floor.
3. Secure the suction hose to the tank frame using a strap or short rope (prevents the strainer from flipping upside down when the pump pulls a vacuum).
4. Connect hard suction to the pumper's master intake.

PHASE 3: RAPID TANKER DUMP & JET SIPHON (25 MINUTES) — TIMED / STRESS

1. **Guiding the Tanker to the Dump Site:**
   - Spotter directs the tanker driver using standard hand signals.
   - Position the tanker's 10-inch dump chute directly over the lip of the drop tank.
   - Tanker driver sets air brake and trips the air-actuated or manual dump valve.
   - Water dumps in a high-volume torrent (aim for 1,500–2,000 Imperial gallons offloaded in 60–90 seconds).
2. **Dual-Tank Jet Siphon Setup (For Continuous Flow):**
   - When setting up a second drop tank for high-flow shuttle operations, place Tank 2 end-to-end with Tank 1.
   - Place a **Jet Siphon** tube between Tank 2 (dump tank) and Tank 1 (drafting tank).
   - Connect a 1.75" hose line from the pumper discharge to the jet siphon inlet.
   - Flowing 50–75 PSI into the jet siphon creates a strong Venturi vacuum, transferring water continuously from Tank 2 into Tank 1 at over 500 GPM!

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Drain sleeve untied:* Dumping 2,000 gallons into a tank with the drain sleeve untied means half your water runs down the ditch!
- ❌ *Setting tank on steep road camber:* If the road crown is too steep, water will spill over the downhill rail before the tank is even 60% full.
- ❌ *Strainer vortexing:* At low water levels (under 6 inches), swirling air vortexes form. Keep a floating ball, plastic lid, or rubber mat over the strainer inlet to break up whirlpools.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How many seconds did it take from tanker stopping to dump valve open?", "Did the jet siphon keep Tank 1 level with Tank 2?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Drain sleeve opened to release remaining water.
- ☐ Liner wiped dry and swept clear of gravel/mud before folding.
- ☐ Frame folded neatly and secured back in the apparatus mounting bracket.
- ☐ Suction hose and strainer drained and mounted.
- ☐ Tanker refilled 100% from station fill line.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 11

RURAL WATER SUPPLY

NFPA 1001 Level 1 (JPR 4.3.15) / NFPA 1002

DRAFTING FROM A STATIC WATER SOURCE & DRY HYDRANT

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Gear or Work Boots/Gloves (PFD if near water edge)

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Pump Operator & Suction Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to assemble 10 to 20 feet of hard suction hose with a strainer, connect to the pumper's master intake, submerge the strainer into a static water source (pond, river, stream, cistern) or couple to a municipal **dry hydrant**, pull a continuous prime within **30 seconds**, and transition from tank water to draft without cavitation. The foundation of rural firefighting.

📦 TOOLS & APPARATUS SETUP

- Pumper with 2 sections of 6-inch (150mm) hard suction hose and master intake.
- Barrel strainer (with tie-off rope) and floating dock/strainer or dry hydrant adapter.
- Static water source: Station cistern, dry hydrant, local pond, brook, or drafting pit.
- Personal Flotation Devices (PFDs) for any members working within 3 metres (10 feet) of open water.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Water Edge Safety:** Any member working within 10 feet of a steep riverbank, pond edge, or ice shelf **must wear a PFD** (life jacket). Never wear full structural turnout gear onto untested winter ice.

⚠️ **Vacuum Gauge Monitoring:** Monitor compound gauge during priming. If vacuum exceeds 22 inches of mercury (inHg) or fluctuates wildly, cavitation is occurring.

🛑 **THE STOP RULE: Shout "STOP!" immediately if bank soil collapses near the water edge, or if a hard suction hose slips off a coupling.**

PHASE 1: HARD SUCTION COUPLING & STRAINER SELECTION (15 MINUTES) — SLOW DRY RUN

1. **Coupling Hard Suction:**
   - Two firefighters grasp the 10-foot hard suction section.
   - Inspect the thick rubber gasket in the female swivel coupling for dirt or tears.
   - Thread coupling onto the pumper's master intake and torque tight with rubber mallet or spanner wrenches (an air leak as small as a pinhole will prevent pulling a prime!).
2. **Selecting & Rigging the Strainer:**
   - **Barrel Strainer:** For deep water (>4 feet). Requires minimum 2 feet of water above the strainer to prevent surface whirlpools, and 1 foot of clearance from muddy bottom. Tie a rope around the strainer eyelet to control depth.
   - **Floating Strainer:** For shallow streams or ponds (12 to 24 inches deep). Floats on surface and drafts from bottom.
   - **Dry Hydrant Connection:** Couple hard suction directly to the 4.5" or 6" dry hydrant threaded/Storz head.

DRILL 11: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Rural Water Supply • Practical Application

PHASE 2: PULLING THE PRIME (20 MINUTES) — BUDDY COACHING

1. **Pre-Priming Pump Setup:**
   - Shift transmission into **PUMP GEAR** (confirm speedometer reads ~15–20 mph or green "Pump Engaged" light illuminates).
   - Confirm **Tank-to-Pump** valve is **CLOSED** (vital: an open tank valve allows tank water to satisfy the pump, preventing vacuum from reaching the suction hose).
   - Confirm all discharge valves, bleeders, and drain valves are sealed 100% airtight.
2. **Activating the Primer:**
   - Set engine throttle to recommended priming RPM (typically 1,000–1,200 RPM).
   - Pull and hold the **Primer Control Valve**.
   - Observe the **Compound (Intake) Gauge**: Needle should swing counter-clockwise into vacuum, pulling 10 to 20 inches of mercury (inHg).
   - Listen for tone of primer to change from high-pitched whine to deep rumble as water hits the primer vanes.
   - Watch for a steady stream of water discharging from the primer exhaust tube beneath the truck.
   - Once prime is established (steady discharge), release the primer valve smoothly.
   - Master pressure gauge will immediately climb to 50–100 PSI.

PHASE 3: TRANSITION & MAINTAINING CONTINUOUS FLOW (20 MINUTES) — TIMED / STRESS

1. Open a discharge line (or deck gun / crosslay directed back into the water source) slowly.
2. Adjust engine throttle to maintain desired discharge pressure (100–120 PSI).
3. Open the **Pump Cooler / Recirculating Line** slightly to keep pump from overheating if no water is flowing through nozzles.
4. Practice identifying **Cavitation**: Gravelly rattle in the pump casing and needle bouncing on the discharge gauge. Fix immediately by throttling down or reducing flow.

⚠ COMMON PITFALLS & FIREGROUND HAZARDS

- ✗ *Leaving a bleeder valve cracked open:* Even a 1/8-inch drain valve left open admits enough atmospheric air to destroy the pump's vacuum!
- ✗ *Resting strainer in silt/mud:* Sucking mud, leaves, or stones into the pump clogs impellers and burns out shaft seals. Always suspend the strainer using a tie-off rope.
- ✗ *Forgetting winter ice buildup:* In NB winters, dry hydrants can freeze internally from ground frost. Always check dry hydrant barrel with a weighted plumb line before connecting.

🗣 5-MINUTE TAILGATE DEBRIEF

1. ***"How many seconds did it take to achieve prime?", "Did we hear any air hissing around the hard suction coupling gaskets?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Pump drained: Open master pump drain to clear all raw pond water.
- ☐ Flush pump with clean water from onboard booster tank if brackish/muddy water was drafted.
- ☐ Hard suction disconnected, drained, and secured back in brackets.
- ☐ Strainer cleaned of weeds/debris.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 12

SEARCH & RESCUE

NFPA 1001 Level 1 (JPR 4.3.9) / WorkSafeNB Reg 91-191

PRIMARY SEARCH TECHNIQUES & ORIENTED SEARCH

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Rescue | 🦺 **PPE:** Full Structural Turnout Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | 2-Firefighter Search Teams

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to conduct a rapid, disciplined **Primary Search** in an unfamiliar structure under zero-visibility conditions, maintaining continuous orientation using either a perimeter **Wall Search (Left/Right Hand)** or a **Tool-Assisted Oriented Search**, locating a victim and communicating findings without becoming disoriented.

📦 TOOLS & APPARATUS SETUP

- Apparatus bay, meeting room, or storage area configured with simulated residential furniture (chairs, tables, pallets, mattresses).
- Rescue mannequin or weighted dummy (50–150 lbs).
- Flash hoods reversed over SCBA facepieces to simulate zero visibility (smoke).
- Hand tools: Halligan bar, 6-ft pike pole / NY hook, flashlight, Thermal Imaging Camera (TIC).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **WorkSafeNB 51.6(2) Compliance:** No firefighter searches alone. Partners must maintain voice, visual, or physical contact at all times.

⚠️ **Trip & Snag Hazards:** Searchers must stay low (crawling or tripod position). Watch for head-strikes against overhead framing or tables.

🛑 **THE STOP RULE: Shout "STOP!" immediately if searchers lose orientation, if a member experiences real panic, or if someone steps off a simulated ledge/stairwell.**

PHASE 1: SEARCH PHILOSOPHY & TECHNIQUES (10 MINUTES) — SLOW DRY RUN

1. **The Tripod / Crawl Position:**
   - One knee on the floor, one foot forward with boot out in front, head up, reaching forward with hand/tool.
   - Feeling the floor ahead before putting weight down prevents falling through burnt floorboards or down open stairwells!
2. **Oriented Search vs Perimeter Search:**
   - **Wall Search:** One partner maintains hand/foot contact with the exterior wall (Right-Hand or Left-Hand pattern), sweeping room perimeter and beds.
   - **Oriented Search (Preferred Modern Method):**
   - **Oriented Firefighter (Team Lead):** Stays at the doorway of the room with the TIC and flashlight. Monitors air, hallway conditions, and orientation.
   - **Searcher:** Moves into the room, sweeps the space using a hand tool to extend reach by 6 feet, sweeps around and under beds, returns directly to the oriented firefighter's voice and light at the door.

DRILL 12: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF Search & Rescue • Practical Application

PHASE 2: SLOW WALKTHROUGH & SOUNDING THE FLOOR (15 MINUTES) — BUDDY COACHING

1. Pair teams up.
2. Practice entering a room:
   - Sweep the threshold immediately inside the door (victims often collapse right behind doors!).
   - Chock the door open so it cannot latch behind you.
   - Sweep sweeping arcs with the halligan or pike pole handle across the floor, under furniture, and along walls.
   - Listen periodically: Pause for 3 seconds, call out *"FIRE DEPARTMENT! ANYONE HERE?"*, and listen for groans, coughing, or tapping.

PHASE 3: ZERO-VISIBILITY SCENARIO (25 MINUTES) — TIMED / STRESS

1. Reverse flash hoods over facepieces. Search teams start at the exterior "doorway".
2. Team must:
   - Communicate search pattern: *"Engine 1 entering, conducting Left-Hand search!"*
   - Search room systematically, locating the simulated victim.
   - When victim is found, call out loudly: *"VICTIM FOUND! BEDROOM 1!"*
   - Formulate extraction plan (which exit is closest).
   - Drag victim back along the wall to the entry door.
3. Debrief time elapsed and air consumed (PSI check).

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Leaving the wall without a reference tool:* If both searchers drift into the center of a smoke-filled room, disorientation occurs in seconds. Stay anchored to a wall, line, or your oriented partner!
- ❌ *Passing doors without checking:* Skipping a closed door leaves a fire or a trapped victim behind you.
- ❌ *Searching too fast / missing under beds:* Children and pets hide in closets, under beds, and behind curtains. Sweep completely under furniture before moving past.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the team maintain communication during the search?", "How much air did you consume during 10 minutes of crawling?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Mannequins and furniture returned to staging areas.
- ☐ Hand tools wiped clean and stowed in apparatus mounts.
- ☐ SCBA facepieces cleaned, disinfected, and dried.
- ☐ Spare cylinders refilled.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 13

SEARCH & RESCUE

NFPA 1001 Level 1 (JPR 4.3.9) / WorkSafeNB Reg 91-191

VICTIM DRAGS, CARRIES & WINDOW REMOVALS

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Rescue | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | 2-Firefighter Rescue Teams

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to rapidly extricate an unconscious or incapacitated victim from a structure using the **Webbing Drag**, **Bunker Coat/Collar Drag**, **Two-Person Extremities Carry**, and remove a victim through a standard ground-floor residential window. Moving dead weight with gloved hands is an exhausting physical skill that requires exact leverage and team coordination.

📦 TOOLS & APPARATUS SETUP

- 150-lb rescue mannequin or weighted dummy (or paired firefighters taking turns as "patients").
- 20-ft section of 1-inch tubular webbing tied into a continuous loop (water knot) for each member.
- Station window prop, doorway threshold, or low ladder setup.
- Exercise mats or clear floor area.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Back & Spinal Ergonomics:** Rescuers must lift with their legs and keep their backs straight. No twisting under heavy spinal loads!

⚠️ **Head & Neck Support:** Protect the victim's head from bouncing or dragging against door saddles, concrete stairs, or rough asphalt.

🛑 **THE STOP RULE: Shout "STOP!" immediately if a rescuer slips, feels acute lumbar strain, or if a live participant acting as a victim is pinched or dropped.**

PHASE 1: SOLO RESCUER DRAGS (15 MINUTES) — SLOW DRY RUN

1. **The Webbing Drag (Girth Hitch Method):**
   - Loop a 20-ft continuous webbing strap beneath the victim's back/armpits.
   - Pass the ends through the center loop to create a snug chest harness (girth hitch) around the torso.
   - The rescuer turns forward, places the webbing loop over their shoulder or across their chest, and leans forward to drag the victim using leg drive. Rescuer's hands remain free to hold a flashlight or feel walls!
2. **The Bunker Coat / Shirt Drag:**
   - Used when webbing is not immediately in hand.
   - Cross victim's arms across their chest.
   - Rescuer kneels behind victim's head, grasps the heavy fabric of the coat/collar behind the neck, lifts victim's head and shoulders onto rescuer's thighs to protect the skull, and walks backward.

DRILL 13: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF Search & Rescue • Practical Application

PHASE 2: TWO-PERSON EXTREMITIES CARRY (15 MINUTES) — BUDDY COACHING

1. **Rescuer 1 (Torso / Head):**
   - Kneels behind victim, lifts torso to sitting position.
   - Reaches both arms under victim's armpits and grasps victim's crossed wrists firmly.
2. **Rescuer 2 (Legs):**
   - Kneels between or alongside victim's legs facing the direction of travel (or facing rescuer).
   - Slips arms under victim's knees.
3. **The Lift & Move:**
   - Rescuer 1 calls cadence: *"Ready to lift... 1, 2, 3, LIFT!"*
   - Both lift using leg muscles, keeping victim horizontal, and carry down hallway to exterior doorway.

PHASE 3: GROUND-FLOOR WINDOW REMOVAL (20 MINUTES) — TIMED / STRESS

1. **The Window Siphon:**
   - Interior crew maneuvers victim to the windowsill.
   - Rescuer 1 lifts victim's torso and drapes victim's chest and arms over the windowsill facing outside.
   - Exterior firefighter(s) grasp victim under armpits/arms, pulling torso outward and downward over the sill.
   - Interior firefighter supports victim's legs and guides them over the sill, preventing snagging.
   - Lower victim gently onto a ground tarp outside.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Pulling by wrists or ankles without supporting head:* Dragging a relaxed unconscious victim by the feet allows their head to bounce violently across floors and door thresholds, causing secondary trauma.
- ❌ *Bending at the waist during lifts:* Catastrophic back strain occurs when twisting while bent over. Keep hips square and drive with quads and glutes.
- ❌ *Losing grip due to sweaty/wet gloves:* Bare hands on sweaty skin will slip; use clothing, belts, or webbing straps to secure a mechanical grip.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Which drag allowed you to move the fastest with the least effort?", "How much easier was the webbing harness compared to dragging by clothes?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Webbing straps inspected, rolled, and returned to bunker pants pockets.
- ☐ Rescue mannequins stowed.
- ☐ Gear wiped down.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 14

FORCIBLE ENTRY

NFPA 1001 Level 1 (JPR 4.3.4) / WorkSafeNB Reg 91-191

CONVENTIONAL FORCIBLE ENTRY ("THE IRONS")

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Truck | 🦺 **PPE:** Full Structural Turnout Gear, Eye Protection & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Halligan Firefighter & Axe/Striking Firefighter

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to perform conventional forcible entry on residential and commercial doors using **"The Irons"** (Flathead Axe and Halligan Bar). The two-firefighter team must size up the door, determine swing direction, execute the 3-step sequence (**Gap, Set, Force**), and maintain door control. In rural areas, rapid entry without destroying unnecessary property saves minutes when seconds count.

📦 TOOLS & APPARATUS SETUP

- Forcible entry training door prop (or pallet/scrap 2x4 door simulator).
- 1 set of "The Irons" (8-lb or 6-lb flathead axe and 30-inch Halligan bar).
- Wooden door wedges / chocks.
- Eye protection (safety glasses mandatory under helmet visors).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Flying Debris & Glancing Blows:** The striking firefighter must swing the flathead axe with short, controlled, perpendicular strikes onto the Halligan strike plate. No wild baseball swings!

⚠️ **Hand Placement:** The Halligan firefighter must keep their hands wrapped around the shaft or fork—never curled over the strike plate or between the tool and door frame!

🛑 **THE STOP RULE: Shout "STOP!" immediately if the axe head glances off the Halligan, if hands stray into the strike zone, or if the door pops unexpectedly.**

PHASE 1: SIZE-UP & "TRY BEFORE YOU PRY" (10 MINUTES) — SLOW DRY RUN

1. **The Cardinal Rule: TRY BEFORE YOU PRY!** Rotate the knob and push/pull. You look foolish forcing an unlocked door!
2. **Door Construction & Swing Direction:**
   - **Hinges Visible:** Door swings **OUTWARD** toward you (common in commercial buildings, exterior storage rooms).
   - **No Hinges Visible (Frame Stop Visible):** Door swings **INWARD** away from you (standard residential front/rear doors).
3. **Check for Locks:** Look for deadbolts, rim locks, keyways, and feel door resistance with the halligan adz.

PHASE 2: INWARD-SWINGING DOOR (GAP, SET, FORCE) (20 MINUTES) — BUDDY COACHING

1. **GAP:**
   - Place the bevel of the Halligan **fork** (or adz) against the door edge, 6 inches above or below the lock cylinder.
   - Push the tool into the door stop to create an initial purchase gap between the door and the jamb.
2. **SET:**
   - Halligan firefighter works the fork into the gap, wrapping the curve around the door edge.
   - Halligan firefighter directs the striking firefighter with clear, single-word commands:
   - *"HIT!"* = One crisp, controlled strike on the adz/strike plate.
   - *"HIT!"* = Second strike driving the fork deeper.
   - *"STOP!"* = Cease striking immediately.
   - The fork is "set" when the arch of the fork straddles the inner door jamb.
3. **FORCE:**
   - Halligan firefighter pushes the Halligan bar hard toward the door face (using maximum mechanical leverage).
   - The lock bolt pops past the strike plate, and the door swings inward.
   - **Door Control:** Immediately slip a boot or the axe handle against the door to keep it from swinging wide open and feeding fresh oxygen to an interior fire!

PHASE 3: OUTWARD-SWINGING DOOR (15 MINUTES) — TIMED / STRESS

1. Outward swinging doors are the easiest and fastest to force.
2. Insert the **adz end** of the Halligan between the door and the frame near the lock.
3. Command: *"HIT!"* to drive the adz deep behind the door stop.
4. Push the shaft of the Halligan firmly outward (away from the building). The door pops open toward you instantly.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ✗ *Curling fingers over the Halligan adz:* When the axe strikes, a missed blow or vibration will crush fingers. Keep hands strictly on the center knurling of the Halligan bar.
- ✗ *Swinging the axe like a sledgehammer:* Huge, wide swings miss the target and risk hitting your partner. Short, powerful, 12-to-18-inch strikes from the shoulder offer 10x more control.
- ✗ *Losing door control after the pop:* If you pop the door and let it blow open, modern fire dynamics show flow path acceleration can cause a flashover within 60–90 seconds! Chock or control the door.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Were the 'HIT' and 'STOP' verbal commands crisp and loud?", "Did the Halligan firefighter keep their hands out of the strike path?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Axe blade inspected for chips or burrs; file smooth if needed.
- ☐ Halligan fork and adz checked for bends or cracks.
- ☐ Married together using standard strap or friction fit and returned to apparatus bracket.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 15
VENTILATION

NFPA 1001 Level 1 (JPR 4.3.11, 4.3.12) / WorkSafeNB Reg 91-191

POSITIVE PRESSURE VENTILATION (PPV) & HYDRAULIC VENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Rescue | 🦺 **PPE:** Full Structural Turnout Gear, Eye & Hearing Protection

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Ventilation Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must understand how to position and operate a **Positive Pressure Ventilation (PPV)** fan (gas-powered or electric battery) to establish a "cone of air" covering the entry doorway, coordinate with an exhaust opening to rapidly clear smoke and toxic heat, and execute **Hydraulic Ventilation** using a wide fog stream from an interior attack line.

📦 TOOLS & APPARATUS SETUP

- Apparatus PPV fan (gasoline or electric PPV).
- Attack hose line (1.75" / 45mm) with combination fog nozzle.
- Station bay doors, open hallway, or training structure.
- Hearing protection and eye protection.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Carbon Monoxide (CO) Hazard:** Gasoline-powered fans generate high concentrations of CO in their exhaust. Never point a gas PPV fan into a structure without an open exhaust opening, and monitor interior air with a 4-gas detector.

⚠️ **Hearing Protection:** PPV fans generate high decibel noise. Wear earplugs or headsets during operation.

🛑 **THE STOP RULE: Shout "STOP!" immediately if an uncoordinated fan is energized before interior attack crews confirm water is on the fire, or if smoke starts churning backwards out the intake.**

PHASE 1: PPV PRINCIPLES & THE "CONE OF AIR" (15 MINUTES) — SLOW DRY RUN

1. **The 3 Rules of PPV:**
   - **Rule 1: Coordinate with Attack.** Never start a PPV fan during an active interior structural fire until the interior attack team confirms: *"Fire knocked down, ready for ventilation!"* Introducing high-velocity air into an unknocked-down fire causes rapid thermal runaway (blowtorch effect)!
   - **Rule 2: Exhaust Opening First.** An exhaust opening (window or exterior door) in the fire compartment must be opened **BEFORE** the fan is brought to full throttle. The exhaust opening should be approximately the same size or slightly larger than the intake door (1:1 to 1:1.5 ratio).
   - **Rule 3: Seal the Intake ("Cone of Air").** Position the fan 4 to 10 feet back from the doorway. Tilt the fan shroud so the pressurized air stream forms a visible cone covering all four corners of the door frame.
2. **Checking the Seal:** Walk across the doorway with a gloved hand. You should feel pressurized air pushing inward at all four corners, with no smoke or air swirling backward around the edges.

DRILL 15: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ventilation • Practical Application

PHASE 2: ELECTRIC VS GAS FAN OPERATION (15 MINUTES) — BUDDY COACHING

1. **Gas-Powered Fan:**
   - Check fuel level. Set choke, open fuel valve, pull starter cord smoothly.
   - Allow 30 seconds of warm-up before increasing throttle to full RPM.
2. **Battery / Electric Fan:**
   - Verify battery pack charge status.
   - Position at doorway, press power switch, rotate variable speed dial.
   - Highlight benefits: Zero CO exhaust, instant on/off control, quiet operation allowing fireground radio communication.

PHASE 3: HYDRAULIC VENTILATION (INTERIOR FOG STREAM) (20 MINUTES) — TIMED / STRESS

1. **The Principle:** A flowing fog stream moving through an open window creates a low-pressure Venturi effect that pulls thousands of cubic feet of smoke and steam out behind it.
2. **Execution Steps:**
   - Interior nozzle team positions 2 to 3 feet back from an open exterior window.
   - Adjust fog nozzle to a **wide-fog pattern (60 to 90 degrees)**.
   - Direct the stream so the outer edge of the fog pattern covers 85% to 90% of the window opening.
   - Open the nozzle bail fully.
   - Observe the dramatic drop in interior smoke as air and steam are drafted out the window.
   - Flow for 60 seconds, shut down smoothly, and evaluate visibility.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ✗ *Pressurizing without an exhaust:* Turning on a fan with all windows closed creates extreme pressure, forcing superheated smoke and fire into concealed ceiling chases and attics.
- ✗ *Positioning fan too close to the door:* If placed 1 foot from the door, the air cone does not expand enough to cover the upper door corners, allowing smoke to cycle back into the firefighter's face.
- ✗ *Ignoring wind direction:* Always try to ventilate **with the wind**, not against prevailing head-winds.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did you feel any air escaping around the top corners of the doorway?", "How much quieter was the battery fan compared to the gas unit?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Gas fan fuel topped up; fuel petcock valve shut.
- ☐ Electric fan batteries returned to shore-power chargers.
- ☐ Fan blades and guard wiped clean of dust and soot.
- ☐ Hearing protection stowed.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 16

ROPES & RIGGING

NFPA 1001 Level 1 (JPR 4.3.20) / WorkSafeNB Reg 91-191

FIREGROUND KNOTS & TOOL HOISTING

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Truck | 🦺 **PPE:** Structural Gloves or Work Gloves, Helmet

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Ground Rigging Crew & Elevated Haul Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to tie the 4 core fire service knots (**Bowline, Clove Hitch with safety, Figure-8 on a Bight, Half Hitch**) with gloved hands, and rig/hoist an axe, a Halligan bar, a ground ladder, and an attack hoseline up to a second-story window or roof without tools slipping or falling.

📦 TOOLS & APPARATUS SETUP

- 100-ft (30m) utility rope bag (minimum 1/2-inch nylon or polyester kernmantle rope).
- Tools to hoist: Flathead axe, Halligan bar, pike pole, 14-ft roof ladder, 50-ft section of 1.75" hose with nozzle.
- Elevated hoist station (station mezzanine, roof edge, hose tower, or 2nd story window).
- Webbing or edge roller/tarp to protect rope from sharp masonry parapets.

⚠ SAFETY WATCH-OUTS & STOP RULE

⚠ **Drop Zone Safety:** Establish a clear 5-metre (15-foot) **Drop Zone** directly below the hoist opening. No personnel may stand beneath hoisted tools.

⚠ **Tag Line Requirement:** A secondary tag line must be attached to the base of heavy tools or ladders to guide them clear of siding, windows, and obstacles during lifting.

⚠ **Rope Abrasion:** Always place a hose roller, canvas tarp, or turnout coat under the rope at the roof/window edge to prevent sheath damage.

🛑 **THE STOP RULE: Shout "STOP!" immediately if a knot begins to slip, if anyone enters the drop zone, or if the rope rubs bare over a sharp metal flashing.**

PHASE 1: THE CORE 4 KNOTS (15 MINUTES) — SLOW DRY RUN

1. **Bowline:** The premier rescue/loop knot. Forms a secure loop that will not slip or tighten under load.
2. **Clove Hitch:** Forms a tight grip around cylindrical objects (handles, pipes, hose). Always lock with an **Overhand Safety Knot** on the working end.
3. **Figure-8 on a Bight:** Quick, super-strong loop in the end or middle of a rope; easy to untie after heavy loading.
4. **Half Hitch:** Supporting/stabilizing knot used in combination with a clove hitch or bowline to keep long tools vertical while hoisting.

DRILL 16: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ropes & Rigging • Practical Application

PHASE 2: HOISTING TOOLS STEP-BY-STEP (20 MINUTES) — BUDDY COACHING

1. **Hoisting an Axe (Axe Head DOWN):**
   - Tie a **Clove Hitch** (with safety) around the axe handle adjacent to the head.
   - Pull the standing rope along the handle toward the butt.
   - Cast a **Half Hitch** around the end of the handle.
   - Result: When pulled upward, the axe hangs strictly vertical with the heavy head pointing down.
2. **Hoisting a Pike Pole / NY Hook (Head UP):**
   - Tie a **Clove Hitch** (with safety) around the base of the handle.
   - Cast a **Half Hitch** around the center of the handle.
   - Cast a second **Half Hitch** around the hook/head of the tool.
   - Result: Hoisted point-up; hook cannot snag on gutters or window sills.
3. **Hoisting a Ground Ladder:**
   - Tie a **Figure-8 on a Bight** to create a 3-foot loop.
   - Pass the loop through the rungs 1/3 of the way down from the tip, wrap around beams, and loop over the top rungs.
   - Attach a secondary rope as a tag line to the bottom rung.
   - Hoist vertically with tip up.

PHASE 3: HOISTING A CHARGED VS UNCHARGED HOSELINE (15 MINUTES) — TIMED / STRESS

1. **Uncharged Line:** Fold 3–4 feet of hose back onto itself over the nozzle. Tie a **Clove Hitch** with safety around the fold and nozzle, and cast a **Half Hitch** around the hose 2 feet back.
2. **Charged Line:** Tie a **Clove Hitch** around the hose 1 foot behind the nozzle coupling. Pass the bight through the nozzle bail handle and loop over the tip. The rope pulls against the coupling, keeping the nozzle bail clamped closed during the lift.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Forgetting the overhand safety knot:* A clove hitch tied on a wooden or fiberglass tool handle can roll and untie itself under cyclic bouncing without a safety knot!
- ❌ *Hoisting axe with head pointing UP:* If hoisted head-up, the axe blade hooks on window sills, gutters, and siding, creating severe hang-ups.
- ❌ *Failing to use a tag line:* A dangling ladder or halligan swinging in the wind will shatter first-floor windows as it is hauled upward. Always use a ground firefighter on a tag line!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did anyone struggle tying knots while wearing structural gloves?", "Was the drop zone strictly maintained?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Utility ropes inspected along entire length for frays, flat spots, or chemical burns.
- ☐ Rope re-bagged into throw bag (stuffing method, never coiled).
- ☐ Hand tools inspected and remounted on apparatus.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 17

SALVAGE & OVERHAUL

NFPA 1001 Level 1 (JPR 4.3.13, 4.3.14) / WorkSafeNB Reg 91-191

SALVAGE, WATER CHUTES & OVERHAUL WITH TIC

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Rescue | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Salvage Crew Pairs

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to deploy a heavy salvage cover (tarp) using the **Two-Firefighter Balloon Throw**, construct a functional **Water Chute** and **Catch-All Basin** to divert ceiling runoff out of a building, and utilize a **Thermal Imaging Camera (TIC)** alongside pike poles to locate concealed fire in ceilings and wall cavities without causing unnecessary structural destruction.

📦 TOOLS & APPARATUS SETUP

- 2 heavy-duty canvas or vinyl salvage covers (minimum 12' x 14' or 14' x 18').
- 2 pike poles (6-ft fiberglass or wood).
- 1 Thermal Imaging Camera (TIC).
- Simulated furniture pile or tables stacked together.
- Clear water bucket or garden hose for water diversion demonstration.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Overhead Collapse:** During overhaul and ceiling pull, never stand directly underneath the drywall/plaster being pulled. Pull from an angle with your back to an exit!

⚠️ **Eye & Respiratory Protection:** Wear eye protection and keep SCBA on air until the atmosphere has been metered and cleared of toxic particulate and carbon monoxide.

🛑 **THE STOP RULE: Shout "STOP!" immediately if ceiling joists show signs of structural sag or if heavy water buildup threatens catastrophic ceiling collapse.**

PHASE 1: FURNITURE GROUPING & THE BALLOON THROW (15 MINUTES) — SLOW DRY RUN

1. **Grouping Contents:**
   - Move household furniture (dressers, chairs, electronics) to the center of the room, away from exterior walls and light fixtures.
   - Place delicate electronics and photos on top of beds or tables; elevate furniture off wet carpet using blocks.
2. **The Two-Firefighter Balloon Throw:**
   - Two firefighters position on opposite sides of the grouped pile, holding the folded salvage cover.
   - Unfold the cover lengthwise between them.
   - Each firefighter grasps the outer corners with palms facing down.
   - On count: *"1, 2, 3, LIFT!"* Both firefighters make a high, sweeping upward motion, capturing a "balloon" of air beneath the tarp.
   - Guide the floating tarp smoothly over the top of the furniture stack and tuck the bottom edges under the furniture legs to shed water.

DRILL 17: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Salvage & Overhaul • Practical Application

PHASE 2: CONSTRUCTING A WATER CHUTE (15 MINUTES) — BUDDY COACHING

(Used when water is leaking through an upper floor ceiling and must be diverted out a window or doorway)

1. Lay a salvage cover flat on the floor.
2. Place two 6-foot pike poles along the outer edges of the tarp, parallel to each other.
3. Roll the sides of the tarp tightly inward around the pike poles 3 to 4 turns, creating a heavy-duty trough with 6-inch side walls.
4. Position the elevated end beneath the ceiling drip, propped on a ladder, cabinet, or held by a pike pole prop.
5. Direct the lower downhill end of the trough out of an open window or exterior door.
6. Test by pouring a bucket of water down the chute; confirm 100% water diversion without side spills.

PHASE 3: CONSTRUCTING A CATCH-ALL BASIN (10 MINUTES) — TIMED / STRESS

1. Lay salvage cover flat on the floor beneath an overhead leak.
2. Fold each edge inward approximately 3 feet.
3. Fold the corners at 45-degree angles to create interlocking side walls.
4. Result: A waterproof basin that can hold 100 to 200 gallons of dirty runoff, protecting hardwood floors and carpets.

Phase 4: Overhaul with the Thermal Imaging Camera (TIC) (15 Minutes)

5. **Operating the TIC:**
   - Power on TIC; verify battery level and temperature scale mode.
   - Scan technique: Ceiling -> Walls -> Floor in a systematic "Z" sweep.
6. **Identifying Concealed Fire:**
   - Explain white-hot vs colorized temperature gradients.
   - Identify thermal bridges, heat signatures around electrical outlets, recessed lighting, and heating ducts.
7. **Coordinated Ceiling Pull:**
   - Firefighter with TIC identifies hot spot in ceiling.
   - Second firefighter positions with 6-ft pike pole with back to exit.
   - Drives pike pole point through drywall with blade parallel to joists.
   - Twists tool 90 degrees across joists and pulls down with body weight, stepping backward to let debris fall safely ahead.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Pulling ceiling directly above your head:* Falling plaster, sheetrock, and insulation will crush your helmet, obscure your vision, or drop hot embers down your coat collar. Always pull away from you!
- ❌ *Using dirty/punctured salvage covers:* A torn tarp simply funnels dirty water onto the homeowner's valuables. Inspect and wash covers after every fire.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the balloon throw clear the pile without snagging?", "How much cleaner is overhaul when directed by a TIC rather than blind pulling?"***

✅ APPARATUS RESET CHECKLIST

- [ ] Salvage covers washed clean of mud and soot, hung to dry, or folded into standard firefighter accordion fold.
- [ ] Pike pole heads cleaned and inspected for cracks.
- [ ] TIC screen wiped with soft cloth; battery clicked into charger.
- [ ] Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 18

RURAL FIREGROUND PRIORITY

NFPA 1001 Level 1 / Rural Operational Standard

CHIMNEY & WOOD STOVE FIRE OPERATIONS

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear, SCBA & Fall Protection

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Interior Crew & Roof Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must master standard tactical protocols for rural **Chimney and Wood Stove Fires**—the highest-frequency winter structural call in New Brunswick. Firefighters must perform a TIC scan of the chimney chase and attic for hidden extension, clear burning fuel from the firebox without spreading smoke, deploy chimney flare sticks / dry chemical / drop chains from the roof, avoid thermal shock, and advise on mandatory **WETT inspection** clearance.

📦 TOOLS & APPARATUS SETUP

- Chimney kit: Metal ash bucket, metal shovel/trowel, chimney flare sticks (Chimfex), plastic baggies with dry chemical ("chimney bombs"), weighted drop-chain with wire brush or chains.
- Thermal Imaging Camera (TIC).
- Charged 1.75" backup attack line stretched to the front door (charged safety line).
- Roof ladder and ground ladder for roof access.
- 4-gas detector (monitoring Carbon Monoxide - CO).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **THE CARDINAL SIN: NEVER spray a solid water hose stream down a hot chimney flue!** Water instantly expands 1,700x into high-pressure steam, causing a steam explosion that shatters clay tile liners, ruptures metal stovepipes, blows burning embers into the living room, and destroys the chimney.

⚠️ **Roof Slip Hazards:** New Brunswick roofs in winter have hidden frost, snow, or ice under chimney soot. Roof ladder deployment is mandatory.

⚠️ **Attic Extension:** A chimney fire burning creosote at >1,000°C (2,000°F) easily conducts radiant heat into adjacent attic wood trusses.

🛑 **THE STOP RULE: Shout "STOP!" immediately if anyone approaches an un-secured roof edge or if a heavy water line is pointed toward an open flue.**

PHASE 1: TREAT AS A FULL STRUCTURE FIRE & SIZE-UP (10 MINUTES) — SLOW DRY RUN

1. In rural NB, never treat a chimney call as a "routine service call". Roll with structural apparatus and stretch a charged 1.75" backup line to the front entrance.
2. **Exterior Assessment:** Check chimney cap for heavy rolling sparks, glowing flue pipe, or roaring sound ("freight train").
3. **Interior TIC Sweep:** Crew enters with TIC and immediately scans:
   - Floor 1: Wall area, mantle, and ceiling surrounding the fireplace/stove.
   - Floor 2 (if present): Chimney chase in bedrooms and closets.
   - Attic: Mandatory inspection of where the chimney penetrates the attic floor and roof decking. Look for orange/white heat glow.

DRILL 18: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Rural Fireground Priority • Practical Application

PHASE 2: INTERIOR FIREBOX OPERATIONS (15 MINUTES) — BUDDY COACHING

1. Close wood stove air intake dampers to suffocate combustion airflow.
2. Put on SCBA facepiece (on air) to protect against intense smoke and CO.
3. Open firebox door slowly (avoid sudden oxygen rush).
4. Using metal shovel/trowel, transfer burning logs and coals into a metal ash bucket.
5. Carry the bucket immediately outside to the gravel driveway and douse with water.
6. **Deploying Extinguishing Flare (Chimfex):**
   - Strike the flare cap. Place the burning flare inside the firebox.
   - Close the stove door tightly. The flare consumes available oxygen and releases fire-suppressing chemicals that travel up the flue, knocking out creosote combustion in 60–90 seconds.

PHASE 3: ROOF FLUE EXTINGUISHMENT (20 MINUTES) — TIMED / STRESS

1. **Accessing the Peak:**
   - Raise 24-ft extension ladder to eaves; deploy 14-ft roof ladder over the roof ridge.
   - Firefighter climbs with safety harness and chimney kit.
2. **Method 1: Dry Chemical "Chimney Bombs":**
   - Take 1/2 cup of dry chemical powder sealed inside a sandwich ziplock bag.
   - Drop the bag directly down the chimney flue from the top.
   - The plastic bag melts halfway down the roaring flue, releasing a fine cloud of dry chemical powder that extinguishes the burning creosote instantly without water damage!
3. **Method 2: Weighted Drop Chains:**
   - If heavy creosote or a chimney blockage remains, lower a weighted link chain down the flue.
   - Gently swing and bounce the chain against the inner walls to break loose burning creosote flakes, which drop into the cleared cleanout/firebox below.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Leaving the scene without checking the attic:* Creosote fires often crack the masonry flue inside the attic where nobody looks. The fire smoulders in wood trusses and flashes over 2 hours after the department leaves. Always verify attic clearance with a TIC!
- ❌ *Not red-tagging the appliance:* Inform the homeowner in writing that the heating appliance is **out of service** and cannot be used until professionally inspected and certified by a **WETT (Wood Energy Technology Transfer)** technician.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the TIC reveal thermal conduction through sheetrock?", "Why must we never discharge a fog nozzle down the flue?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Chimney kit restocked: Shovel, metal bucket cleaned, dry chem bags restocked.
- ☐ Ladders returned to truck racks.
- ☐ 4-gas detector powered down and docked in charger.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 19

NFPA 1001 Level 2 (JPR 5.1.1, 5.1.2) / WorkSafeNB Reg 91-191

INCIDENT COMMAND & COMMUNICATION

FIRST-DUE SIZE-UP & THE CAN RADIO REPORT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Command

🦺 **PPE:** Structural Turnout Gear or Duty Uniform + Portable Radio

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Senior Firefighter / Acting Officer Participants

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter (especially senior firefighters stepping up when officers are absent) must be able to conduct a rapid **360-degree size-up** upon arrival, transmit a professional **Initial Radio Report (IRR)**, enforce **WorkSafeNB 2-In / 2-Out** rules, and deliver structured progress reports using the **CAN (Conditions, Actions, Needs)** format.

📦 TOOLS & APPARATUS SETUP

- Portable handheld radios on station tactical channel (e.g., Tac 1).
- Training scenario flashcards or photos of local rural residential/commercial structures projected on a screen or clipboard.
- Notepad / grease-pencil command board.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **WorkSafeNB Minimum Staffing Law:** Remind all participants that under WorkSafeNB Reg 91-191 Section 51.6(2) and NFPA 1500, a minimum of **4 firefighters on scene** is legally required before entering an IDLH structural fire (2 inside, 2 standby outside). Arriving with 2 or 3 firefighters means operating in **defensive or exterior transitional mode** unless an immediate, confirmed rescue exception applies!

⚠️ **Radio Discipline:** Enforce strict radio etiquette: think before you press the PTT (push-to-talk) button. No rambling.

PHASE 1: THE 360-DEGREE SIZE-UP WALKAROUND (15 MINUTES) — SLOW DRY RUN

1. Arriving on scene, the first-due senior member conducts a **360-degree walkaround** (or visually checks all sides: Alpha, Bravo, Charlie, Delta).
2. **What to Look For:**
   - **Building:** Stories, construction (wood frame, brick, modular), roof type.
   - **Smoke & Fire:** Location, color, volume, velocity (where is it coming from?).
   - **Life Safety:** Occupancy status, cars in driveway, toys in yard, window signs.
   - **Basement / Grade Level:** Walkout basement on Side Charlie (common in NB hills!).
   - **Hazards:** Overhead power lines down, propane tanks, fuel oil tanks.

DRILL 19: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Incident Command & Communication • Practical Application

PHASE 2: THE INITIAL RADIO REPORT (IRR) (15 MINUTES) — BUDDY COACHING

Practice delivering the standardized 5-part radio report:

```
+--------------------------------------------------------------------------+
| 1. Unit ID: "Engine 1 on scene..." |
| 2. Address/Location: "...at 4522 Route 105." |
| 3. Building Desc: "We have a two-story wood-frame residential dwelling." |
| 4. Conditions: "Working fire, heavy black smoke from Side Bravo 1st." |
| 5. Mode & Action: "Engine 1 crew is laying a 1.75 crosslay for an |
| exterior transitional attack. Establishing Main St |
| Command. We are in defensive mode awaiting backup." |
+--------------------------------------------------------------------------+
```

PHASE 3: THE CAN PROGRESS REPORT (20 MINUTES) — TIMED / STRESS

When Incident Command or dispatch calls for an update, use the **CAN** format:

- **C – Conditions:** What do you see right now?
  (e.g., "Heavy smoke layer banking down to knee level; high heat in kitchen.")
- **A – Actions:** What is your crew doing right now?
  (e.g., "Water is flowing on the ceiling; advancing toward the kitchen.")
- **N – Needs:** What resources or help do you need next?
  (e.g., "Need horizontal ventilation at Side Bravo window; need Tanker 2 to dump water.")

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Skipping the 360-degree walk:* A fire that looks small from the front (Side Alpha) may be fully engulfing a walkout basement at the rear (Side Charlie). Never commit crews without seeing the rear!
- ❌ *Entering IDLH with only 2 or 3 members on scene:* Violates provincial safety laws and puts members at extreme risk without a backup rescue team in place.
- ❌ *Rambling on the radio:* Keep transmissions under 10 seconds. Clear, crisp, and standardized.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the radio report clearly tell incoming units what to expect?", "Did we account for the 2-in/2-out rule based on our simulated staffing?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Radios returned to charging banks; verify battery lights turn amber/red (charging).
- ☐ Scenario sheets filed in training binder.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 20

FIREFIGHTER SURVIVAL

NFPA 1001 Level 2 (JPR 5.3.1, 5.3.3) / WorkSafeNB Reg 91-191

MAYDAY TRANSMISSION & FIREFIGHTER SELF-RESCUE

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Rescue | 🦺 **PPE:** Full Structural Turnout Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to recognize a personal fireground emergency, transmit a crisp **MAYDAY** declaration over the radio using the **LUNAR** format, silence their PASS device during transmissions, follow the **GRAB LIVES** survival checklist, and execute a **Reduced Profile SCBA Maneuver** to escape through 16-inch on-center wall studs.

📦 TOOLS & APPARATUS SETUP

- Full SCBA on air for all participants.
- Portable radios on training channel.
- Wall stud prop: 2x4 studs spaced 16 inches on-center (or between two ladders).
- Entanglement prop (simulated wire box: slack wire cables, BX wire, Romex strung in a frame).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Mask Seal Integrity:** During the reduced profile maneuver, watch that facepieces are not twisted, dislodged, or pulled off by wall studs or wire snags.

⚠️ **Panic Management:** If a member becomes stuck in the entanglement prop, do not let them flail. Coach them to breathe calmly.

🛑 **THE STOP RULE: Shout "STOP!" immediately if an SCBA regulator uncouples or if a member experiences true airway compromise.**

PHASE 1: CALLING THE MAYDAY & THE LUNAR REPORT (15 MINUTES) — SLOW DRY RUN

1. **When to Call a Mayday:**
   - Lost, disoriented, or separated from your crew.
   - Trapped by structural collapse or tangled in wiring.
   - SCBA failure, out of air, or low-air alarm sounding without an exit found.
   - Fall through floor or roof.
2. **The Transmission Protocol:**
   - Depress radio PTT button: *"MAYDAY, MAYDAY, MAYDAY! Engine 1 Interior with a Mayday!"*
   - Await IC acknowledgment: *"Command copies Mayday, go with your Mayday."*
   - Deliver the **LUNAR** report:
   - **L – Location:** *"Division 2, Side Bravo, rear bedroom."*
   - **U – Unit:** *"Engine 1 Attack Team."*
   - **N – Name:** *"Firefighter Smith."*
   - **A – Air / Assignment:** *"Search team, 1,200 PSI remaining."*
   - **R – Resources Needed:** *"Ceiling collapse, pinned under joists, need RIT and irons."*

DRILL 20: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Firefighter Survival • Practical Application

PHASE 2: THE GRAB LIVES SURVIVAL ACTIONS (15 MINUTES) — BUDDY COACHING

- **G – Gauge:** Check cylinder pressure.
- **R – Radio:** Call Mayday (LUNAR).
- **A – Activate PASS:** Turn on PASS device manually.
- **B – Breathing:** Drop to skip/box breathing to conserve air.
- **L – Low:** Stay on the floor where air is coolest.
- **I – Illuminate:** Point flashlight beam straight up toward the ceiling.
- **V – Volume:** Bang a hand tool against the floor/wall rhythmically (3 taps).
- **E – Exit Search:** Crawl along walls or follow hose line ("Smooth bump-bump to the pump").
- **S – Shield Airway:** Protect regulator and facepiece from heat.

PHASE 3: THE REDUCED PROFILE MANEUVER (16" STUDS) (20 MINUTES) — TIMED / STRESS

1. Interior firefighter trapped in a room with no exit except an interior partition wall.
2. Breach drywall with Halligan or axe between two 16" studs; clear sheetrock from both sides.
3. Loosen your **right shoulder strap** on the SCBA harness completely.
4. Unbuckle your **waist belt** and **chest strap**.
5. Slip your right arm out of the harness strap. Swing the SCBA pack off your back, holding the left shoulder strap firmly with your left hand.
6. Push the SCBA cylinder through the 16" studs ahead of you (or hold it along your left ribcage).
7. Turn your body sideways, duck your head through, and slide through the studs like a swimmer.
8. Once clear, slip right arm back in, tighten shoulder strap and waist belt, and verify facepiece seal!

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Waiting too long to call Mayday:* Waiting until your low-air bell rings before calling Mayday gives the RIT team only 2–3 minutes to reach you. Call early!
- ❌ *Screaming over your PASS alarm:* The PASS alarm must be temporarily silenced while pressing the PTT button to transmit LUNAR, then re-activated immediately.
- ❌ *Disconnecting regulator from facepiece:* Never remove your regulator inside a toxic atmosphere, even if air is running low!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How long did it take to deliver the LUNAR report under stress?", "Did anyone snag their regulator hose during the stud crawl?"***

✅ APPARATUS RESET CHECKLIST

- ☐ SCBA harnesses inspected for torn stitching or bent buckles.
- ☐ PASS devices reset and switched off.
- ☐ Cylinders refilled to 4,500 PSI.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 21

RAPID INTERVENTION (RIT)

NFPA 1001 Level 2 (JPR 5.3.3) / WorkSafeNB Reg 91-191

DOWNED FIREFIGHTER PACKAGING & FAST / RIT BASICS

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Rescue / Engine | 🦺 **PPE:** Full Structural Turnout Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | 2-Firefighter RIT Team & Downed Firefighter

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter serving on an exterior standby or Rapid Intervention Team (RIT/FAST) must be able to locate a downed firefighter in zero visibility, assess their air status, transfill high-pressure air via the **Universal Air Connection (UAC)**, convert their SCBA harness into a drag harness (or deploy the built-in **DRD - Drag Rescue Device**), and drag them to safety up or down stairs.

📦 TOOLS & APPARATUS SETUP

- RIT Pack (60-minute air cylinder, high-pressure UAC transfill hose, low-pressure EBSS hose, spare facepiece).
- 1 firefighter in full gear/SCBA acting as "Downed Firefighter" (or 180-lb mannequin).
- 20-ft length of 1-inch tubular webbing.
- Flash hoods reversed over facepieces to simulate zero visibility.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Facepiece Seal Security:** When dragging a downed firefighter, watch that their facepiece is not pulled off by rubbing against floors or stairs.

⚠️ **High-Pressure Equalization:** When connecting the UAC transfill whip, listen for the rapid rush of equalizing air (can take 30–60 seconds to balance cylinders).

🛑 **THE STOP RULE: Shout "STOP!" immediately if the downed firefighter's airway or regulator hose is pinched or disconnected during dragging.**

PHASE 1: RAPID ASSESSMENT UPON REACHING DOWNED FIREFIGHTER (10 MINUTES) — SLOW DRY RUN

1. Team arrives at downed firefighter.
2. Turn off downed firefighter's PASS device to allow verbal communication.
3. **Air Assessment (The 3-Step Air Check):**
   - Feel for air exhausting from the regulator (is member breathing?).
   - Check remote pressure gauge (is cylinder empty?).
   - Confirm facepiece is seated firmly against face.

DRILL 21: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Rapid Intervention (RIT) • Practical Application

PHASE 2: HIGH-PRESSURE AIR TRANSFILL (UAC) (15 MINUTES) — BUDDY COACHING

1. If the downed firefighter is breathing but low on air:
   - Deploy the high-pressure transfill whip from the RIT pack.
   - Locate the **Universal Air Connection (UAC)** port on the downed firefighter's SCBA harness (located within 4 inches of the main cylinder valve).
   - Pop the rubber dust cap off the UAC port.
   - Push the transfill female coupling straight onto the UAC male nipple until it clicks and locks into place.
   - High-pressure air immediately transfers and equalizes between the RIT cylinder and the downed firefighter's cylinder.
   - Do not disconnect until pressure equalizes (gauge reads balanced).

PHASE 3: PACKAGING & THE SCBA HARNESS CONVERSION (20 MINUTES) — TIMED / STRESS

1. **The Harness Leg-Loop Conversion:**
   - Downed firefighters slide out of their SCBA harness if dragged by the shoulder straps!
   - Unbuckle the waist belt.
   - Pull the waist belt buckle down between the downed firefighter's legs.
   - Re-buckle the waist strap around the groin/thighs.
   - Result: The SCBA harness is now a full-body climbing harness; pulling on the shoulder straps or cylinder will not pull the pack off their back!
2. **Deploying the DRD (Drag Rescue Device):**
   - Grasp the Velcro flap at the back of the bunker coat collar.
   - Pull the built-in webbing loop out until fully extended (creates a 2-foot drag handle around the firefighter's upper torso).
3. **Staircase Extrication (Head First):**
   - When dragging a downed firefighter up stairs, position rescuer facing backwards up the stairs.
   - Grasp the converted harness straps or DRD handle.
   - Rest the downed firefighter's head and neck on your boots/shins to prevent their skull from bouncing on stair treads.
   - Walk backward up the stairs using leg drive.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Dragging without converting the harness:* Dragging a firefighter by their coat or unbuckled pack will slide the coat over their head and rip off their facepiece, exposing their airway to superheated gases!
- ❌ *Fumbling for the UAC port:* In zero visibility, feel for the cylinder handwheel; the UAC port is located right beside it. Practice finding it blindfolded.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How long did it take to locate the UAC port blindfolded?", "Did the converted waist belt prevent the pack from sliding off?"***

✅ APPARATUS RESET CHECKLIST

- ☐ RIT pack cylinder refilled to 4,500 PSI; transfill hose wiped and re-bagged.
- ☐ Turnout coat DRD straps pushed back into collar flaps and velcroed flat.
- ☐ SCBA harness straps untangled and restored to normal donning position.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 22

VEHICLE EXTRICATION

NFPA 1001 Level 2 (JPR 5.4.1) / Rural Rescue

VEHICLE STABILIZATION & GLASS MANAGEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Rescue / Engine

🦺 **PPE:** Extrication Gloves, Bunker Gear, Eye & Respiratory Protection

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Stabilization & Glass Crews

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to stabilize a vehicle on its wheels or side using **Step Chocks**, **Box Cribbing**, and **Rescue Struts**, chock the wheels to prevent roll, and safely manage **Tempered vs Laminated Glass** without spraying glass fragments over trapped patients. Unstable vehicles and uncontrolled glass cause severe responder and patient injuries at rural highway collisions.

📦 TOOLS & APPARATUS SETUP

- Training vehicle (scrap car, municipal pickup, or mock frame).
- 4 step chocks, 12–16 pieces of 4x4 box cribbing, 4 wedges.
- Mechanical or ratchet rescue struts (if equipped).
- Glass tools: Spring-loaded center punch, glass master / reciprocating saw, heavy duct tape or contact paper, canvas tarp / patient blanket.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Vehicle Suspension Movement:** Never place hands, arms, or boots underneath a vehicle's rocker panels or tires until stabilization is completed.

⚠️ **Glass Dust Inhalation:** Modern windshields (laminated glass) produce fine, hazardous glass dust when cut. Rescuers and patients must wear N95/particulate masks and safety glasses.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the vehicle rocks, shifts, or slips from cribbing while personnel are working around it.**

PHASE 1: 4-POINT VEHICLE STABILIZATION (WHEELS ON GROUND) (15 MINUTES) — SLOW DRY RUN

1. **Prevent Roll:** Place rubber or wooden wheel chocks tightly in front of and behind the uninjured rear tire.
2. **Defeat the Suspension (Step Chocking):**
   - A car resting solely on its inflated tires and coil springs will bounce violently during extrication tool use.
   - Slide **Step Chocks** beneath the rocker panels at four structural points:
   - Just behind the front wheels (front A-pillar).
   - Just ahead of the rear wheels (rear C-pillar).
   - Insert chocks with the steps facing upward; tap wooden wedges into any gaps until the car frame sits solid on the wood.
   - Deflate tires (pull valve stems) to settle the chassis firmly onto the step chocks.
   - Rock test: Push firmly on the vehicle fender. The vehicle should feel rigid with **zero suspension bounce**.

DRILL 22: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Vehicle Extrication • Practical Application

PHASE 2: BOX CRIBBING (VEHICLE ON SIDE / LIFTED LOAD) (15 MINUTES) — BUDDY COACHING

1. **The 4x4 Box Crib:**
   - Build a stable box using two parallel 4x4 timbers, overlaid perpendicularly with two more 4x4s.
   - Rule of thumb: Crib height must never exceed **3 times the contact width** (e.g., an 18" wide base can only be stacked up to 54" high).
   - Top off with angled wedges to ensure 100% surface contact with the angled frame.

PHASE 3: GLASS MANAGEMENT (TEMPERED VS LAMINATED) (20 MINUTES) — TIMED / STRESS

1. **Identify the Glass:**
   - **Tempered Glass (Side & Rear Windows):** Shatters into thousands of small, blunt cubes when struck.
   - **Laminated Glass (Windshields & Many Modern Side Windows):** Two layers of glass bonded to a plastic sheet; will not shatter with a center punch. Look for the "Laminated" bug/stamp in the corner!
2. **Managing Tempered Glass (Side Windows):**
   - Cover window with adhesive film or cross-strips of heavy duct tape to capture fragments.
   - Cover interior patient with a heavy blanket or salvage tarp.
   - Position a **Spring-Loaded Center Punch** in the **lowest, outermost corner** of the window (never center of glass!).
   - Push until punch clicks. Window shatters into tape. Rake glass fragments outward using a gloved hand or tool.
3. **Managing Laminated Glass (Windshield):**
   - Cut around the perimeter using a manual **Glass Master** or reciprocating saw (fine metal blade, 14–18 TPI).
   - Spray water or shaving cream along the cut line to capture glass dust.
   - Peel the cut windshield outward and discard under the vehicle or in the scrap pile.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Punching tempered glass in the center:* The center of the glass flexes and resists the punch. Always punch the lower corner where glass is stiffest!
- ❌ *Stacking cribbing too high:* Tapering or building a narrow, wobbly tower will kick out when hydraulic spreaders exert 10,000+ lbs of force on the door!
- ❌ *Ignoring patient protection:* Shattering a side window without shielding the driver's face sprays sharp cubes directly into their eyes.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the vehicle bounce after chocks were set?", "How did we distinguish between tempered and laminated side glass?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Glass shards swept clear of training area.
- ☐ Cribbing and step chocks inspected for splits or rot; stacked neatly in truck trays.
- ☐ Center punches and saw blades cleaned and returned to rescue box.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 23

VEHICLE EXTRICATION

NFPA 1001 Level 2 (JPR 5.4.1) / Rural Rescue

HYDRAULIC RESCUE TOOLS – DOOR POP & HINGE REMOVAL

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Rescue / Engine | 🦺 **PPE:** Extrication Gloves, Bunker Gear, Helmet & Safety Glasses

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Tool Operator, Backup Firefighter & Patient Shield

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to safely operate hydraulic extrication tools (power unit, spreaders, and cutters—hydraulic hose or battery e-draulic), create a purchase point on a jammed vehicle door, force the door latch or shear the hinges, and remove the door completely to create a wide patient extrication path within **5 minutes**.

📦 TOOLS & APPARATUS SETUP

- Hydraulic rescue tool set: Spreader, cutter, hydraulic power unit (if hydraulic) or fresh batteries (if battery e-draulic).
- Scrap training vehicle with intact doors.
- Halligan bar (for creating initial gap if needed).
- Patient shield (hard polycarbonate board or heavy turnout blanket).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Tool Reaction & Pinch Zone:** Hydraulic spreaders exert up to 40,000 lbs of force. As the tips spread, the tool body can kick upward or pivot violently into the operator's thighs or ribs. Operators must position their bodies **outside the tool's swing arc**.

⚠️ **Hand Guards:** Keep gloved hands strictly behind the safety hand-guard. Never place fingers near the moving arms or cutting blades!

⚠️ **Pressurized Hose Line:** When operating hydraulic or fuel-powered rescue tools around motor vehicles, a **charged 1.75" dry chemical or foam line** must be staffed and staged as a fire suppression standby.

🛑 **THE STOP RULE: Shout "STOP!" immediately if tool tips slip out of a purchase point, or if the tool body begins to pinch the operator against the car frame.**

PHASE 1: TOOL INSPECTION & POWER UNIT SETUP (10 MINUTES) — SLOW DRY RUN

1. **Hydraulic Hose Connection (Core Coaxial / Dual Hose):**
   - Inspect couplings for grit/metal shavings. Connect and twist locking collar.
   - Start hydraulic gas power unit; verify choke off, engine at operating throttle, and control valve set to "Pressure".
2. **Battery (e-Draulic) Verification:**
   - Insert charged lithium-ion battery until latch clicks; check battery indicator (must show 4 green LEDs).
3. **Control Handle Familiarization:**
   - Modern tools use a deadman twist throttle: rotate right to open/spread, rotate left to close/cut. Releasing the handle automatically locks arms in place.

DRILL 23: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Vehicle Extrication • Practical Application

PHASE 2: CREATING THE PURCHASE POINT (15 MINUTES) — BUDDY COACHING

1. **Method 1 (Pinching the Window Frame / Rocker):**
   - Place spreader tips on the metal door skin near the top latch or bottom corner.
   - Squeeze/pinch the sheet metal inward to buckle the door edge away from the N-pillar, creating a 1-to-2-inch gap.
2. **Method 2 (Mechanical Purchase with Halligan):**
   - Drive the adz of a Halligan bar between the door edge and B-pillar at a 45-degree angle to pry the seam open.

PHASE 3: THE DOOR POP (LATCH SIDE VS HINGE SIDE) (25 MINUTES) — TIMED / STRESS

1. **Popping the Latch (B-Pillar Side):**
   - Insert spreader tips fully into the purchase gap (full tip insertion prevents tips from spitting out!).
   - Rotate handle to spread. Watch the sheet metal tear.
   - Work the tips downward toward the heavy **Nader pin (latch)** mechanism.
   - Spread with the full width of the arms. The steel latch mechanism will snap with a loud "BANG!"
2. **Shearing or Spreading the Hinges (A-Pillar Side):**
   - Once the latch is popped, swing the door wide open.
   - To remove the door completely:
   - **Option A (Cutters):** Place hydraulic cutter blades perpendicular to the door hinges. Cut the upper hinge completely, then cut the lower hinge and wire harness.
   - **Option B (Spreaders):** Insert spreader tips directly into the hinge pocket against the car frame and spread; the hinge bolts will shear cleanly off the chassis.
3. Remove the door completely from the vehicle and carry it to the designated scrap pile (sharp edges taped or covered).

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ✗ *Spreading with only the tip ends:* Using only the very tips of the spreader arms causes the tool to spit out under load with explosive force. Always bury the tips deep into the seam!
- ✗ *Cutting high-strength boron steel at an angle:* Twisting the cutter blades while shearing a modern ultra-high-strength steel B-pillar will crack or shatter the tool blades. Always cut **at a clean 90-degree angle** to the post!
- ✗ *Standing between the tool and the car:* If the spreader tips slip, the 50-lb tool body will kick directly into your chest or groin. Stand to the side!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the spreader tips stay seated without slipping?", "How did the operator position their body relative to the tool kick zone?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Hydraulic power unit shut down; pressure dumped using dump valve before uncoupling hoses.
- ☐ Cutter blades inspected for nicks, cracks, or blade spread.
- ☐ Attendance roster signed.
- ☐ Spreader tips left slightly cracked (1/4 inch open, not touching) to relieve internal hydraulic pressure during thermal changes.
- ☐ Tools wiped clean of motor oil and broken glass.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 24

NFPA 1001 Level 2 (JPR 5.4.1) / NFPA 1072

VEHICLE EXTRICATION / HAZMAT

VEHICLE BATTERY ISOLATION – 12V SYSTEMS & ELECTRIC VEHICLES (EV)

⏱️ **Time:** 45–60 min | 🚒 **Apparatus:** Rescue / Engine | 🦺 **PPE:** Bunker Gear, Safety Glasses & Work/Extrication Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Battery Isolation Team

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to safely locate and isolate conventional **12-Volt vehicle batteries** to disarm airbag and electrical fire hazards, and recognize modern **Electric Vehicle (EV) and Hybrid** safety protocols: locating emergency cut loops/first responder disconnects, identifying **bright orange high-voltage cabling**, and understanding high-voltage thermal runaway hazards.

📦 TOOLS & APPARATUS SETUP

- Training vehicle or modern vehicle (volunteer or municipal truck).
- Hand tool battery isolation kit: 10mm/1/2" ratchet wrench, wire cutters/cable shears, electrical tape.
- Emergency Response Guide (ERG) or modern vehicle extrication field guide app (e.g., Euro NCAP / NFPA EV field guide).
- Thermal Imaging Camera (TIC).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **THE HIGH-VOLTAGE CARDINAL RULE: NEVER CUT, PINCH, OR TOUCH BRIGHT ORANGE CABLES!** High-voltage cables in EVs carry 400 to 800+ Volts DC—instantly fatal.

⚠️ **Airbag Capacitor Bleed Time:** Disconnecting the 12V battery does **not** instantly disarm airbags! Capacitors store backup power for **up to 5 to 10 minutes** after battery isolation. Treat all undeployed airbags as loaded weapons.

🛑 **THE STOP RULE: Shout "STOP!" immediately if anyone points a cutting tool near orange cables, or if an EV battery pack shows signs of off-gassing, hissing, or thermal runaway.**

PHASE 1: CONVENTIONAL 12-VOLT BATTERY ISOLATION (15 MINUTES) — SLOW DRY RUN

1 Why Isolate 12V First?

- Eliminates electrical ignition sources around leaking gasoline/antifreeze.
- Cuts power to undeployed Supplemental Restraint Systems (SRS airbags, seatbelt pretensioners).
- Shuts down high-voltage contactors in electric vehicles (EVs require 12V to energize high-voltage relays!).

2 The Disconnect Sequence:

- Pop hood or trunk (look for auxiliary battery locations in trunk or under rear seats).
- Identify terminals: **Negative (-)** (usually black or bare chassis ground) and **Positive (+)** (usually red or plastic cover).
- **Rule: ALWAYS DISCONNECT NEGATIVE (-) FIRST!**
- If your wrench touches the metal car chassis while loosening the negative terminal, nothing happens.
- If you touch the positive terminal first and hit the chassis with your metal wrench, you cause a violent high-amp short circuit, sparks, and battery explosion!
- If cutting cables: Make **two cuts** removing a 2-inch (5cm) chunk of the negative cable so the springy wire ends cannot accidentally spring back and re-contact!

DRILL 24: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Vehicle Extrication / Hazmat • Practical Application

PHASE 2: ELECTRIC VEHICLE (EV) IDENTIFICATION & HIGH-VOLTAGE SYSTEMS (15 MINUTES) — BUDDY COACHING

1. **Identify the EV/Hybrid:**
   - Badges (Hybrid, EV, PHEV, BEV, Lightning, Tesla, Mach-E).
   - Lack of tailpipes / exhaust system.
   - Exterior charging port doors.
   - Large high-voltage battery casing bolted beneath the floor pan.
2. **Visualizing the Orange Cables:**
   - Locate the inverter, electric drive motors, and high-voltage conduit.
   - Reinforce: All 400V–800V cables are coded in **BRIGHT ORANGE**. They are completely off-limits to extrication tools.

PHASE 3: EV FIRST RESPONDER CUT LOOPS & SERVICE PLUGS (20 MINUTES) — TIMED / STRESS

1. **The Emergency Responder Cut Loop:**
   - Most modern EVs feature a low-voltage **First Responder Cut Loop** or emergency disconnect plug (commonly located under the front hood, near the fuse box, or behind the rear trunk trim).
   - Look for the bright colored tag with a scissors icon or pull-loop.
   - Cutting this low-voltage loop immediately opens the internal high-voltage contactors, trapping high voltage securely inside the sealed battery casing.
2. **EV Fire & Submersion Precautions:**
   - If an EV battery is damaged or burning, water requirements are massive (minimum **10,000 to 20,000 Litres** / 3,000–5,000 gallons) to cool the battery cell casing.
   - Use a TIC to monitor battery casing temperature continuously from a safe distance.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Cutting positive battery terminal first:* A metal tool bridging the positive post to the fender creates a blinding arc flash and molten metal spatter. Always isolate the negative ground first!
- ❌ *Cutting into B-pillars without checking for seatbelt inflators:* Modern B-pillars contain pyrotechnic gas cylinders. Cutting through an inflator causes an explosive metal rupture. Peel plastic trim back to inspect before cutting!
- ❌ *Assuming an EV is off because the engine is silent:* An EV can be fully energized and in gear without making any sound. Always chock wheels and remove the key fob 50+ feet away from the vehicle.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Where is the low-voltage cut loop located on our municipal hybrid/EVs?", "Why must we wait 5–10 minutes after disconnecting 12V before cutting near airbags?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Hand tools (10mm wrenches, cable cutters) wiped clean and stowed in extrication kit.
- ☐ Training vehicle hoods closed.
- ☐ Vehicle emergency guidebooks returned to command cab.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 25

ADVANCED FIRE SUPPRESSION

NFPA 1001 Level 2 (JPR 5.3.1) / NFPA 11

FOAM EDUCTORS & FIREFIGHTING FOAM APPLICATION

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Nozzle Team & Pump Operator

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to properly rig an **In-Line Foam Eductor** to an attack line, set appropriate metering percentages for **Class A (wildland/structure)** and **Class B (flammable liquids)** foam concentrates, understand backpressure limitations, and apply foam using the three classic techniques: **Roll-On, Bank-Down**, and **Rain-Down**.

📦 TOOLS & APPARATUS SETUP

- Apparatus pumper with 1.75" (45mm) discharge.
- 95 GPM (or 125 GPM) in-line foam eductor with metering dial and pick-up tube.
- 150 to 200 feet of 1.75" hose line (maximum allowable length downstream of eductor).
- Matching 95 GPM (or 125 GPM) foam or combination fog nozzle.
- 5-gallon pail of training foam concentrate (or water with dish soap for zero-cost environmental drill).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Foam Concentrate Exposure:** Foam concentrate is irritating to skin and eyes. Wear eye protection and structural gloves while handling concentrate pails and pickup tubes.

⚠️ **Environmental Runoff:** Ensure foam solutions do not drain directly into local salmon rivers, brooks, or storm sewers. Conduct drill over contained asphalt or gravel.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the nozzle stream collapses or if foam concentrate spills uncontrollably.**

PHASE 1: THE PHYSICS OF THE IN-LINE EDUCTOR (15 MINUTES) — SLOW DRY RUN

1. **The Venturi Effect:**
   - Water passing through a restricted orifice creates a low-pressure vacuum that draws foam concentrate up the pick-up tube.
2. **The 4 Rules of Foam Eductor Success (Why Eductors Fail):**
   - **Rule 1: Inlet Pressure:** The pump operator must deliver **200 PSI** to the inlet of the eductor. Any less will not create sufficient vacuum.
   - **Rule 2: Matching Gallonage:** The nozzle GPM **MUST MATCH** the eductor GPM rating! (e.g., A 95 GPM eductor requires a 95 GPM nozzle setting. If the nozzle is set to 60 or 150 GPM, foam will not draft!).
   - **Rule 3: Maximum Hose Length:** Never exceed **200 feet (60m)** of 1.75" hose between the eductor and the nozzle. Excessive downstream friction causes backpressure that kills the Venturi vacuum.
   - **Rule 4: Elevation Limit:** Keep the nozzle within 10 to 15 feet of elevation relative to the eductor.

DRILL 25: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Advanced Fire Suppression • Practical Application

PHASE 2: METERING DIAL SETTINGS (10 MINUTES) — BUDDY COACHING

- **Class A Foam (Wood, Paper, Structural, Brush):**
- Set dial to **0.3% to 0.5%** for initial structural knockdown.
- Set dial to **0.1% to 0.2%** for deep penetration during overhaul.
- **Class B Hydrocarbons (Gasoline, Diesel, Heating Oil):**
- Set dial to **1% or 3%** (check manufacturer pail instructions).
- **Class B Polar Solvents (Ethanol, E85, Alcohol, Brake Fluid):**
- Requires AR-AFFF (Alcohol Resistant foam); set dial to **3% or 6%**.

PHASE 3: THE 3 APPLICATION TECHNIQUES (25 MINUTES) — TIMED / STRESS

1. **Technique 1: The Roll-On Method:**
   - Used on pool/spill fires on flat ground.
   - Direct the foam stream at the ground **just in front of the burning spill**.
   - The foam blanket builds up and rolls forward across the liquid surface, smothering vapors without agitating the fuel.
2. **Technique 2: The Bank-Down Method:**
   - Used when an obstacle or vertical wall is present (e.g., burning tanker trailer, wall, engine compartment).
   - Direct the stream against the vertical wall just above the liquid.
   - Foam bounces off the surface and runs gently down to cover the fuel pool.
3. **Technique 3: The Rain-Down Method:**
   - Used when roll-on or bank-down are obstructed.
   - Direct stream high into the air at a 60-degree angle over the fire.
   - Allow the foam to break apart and fall gently down like snowflakes over the spill surface (never plunge stream directly into fuel!).

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Plunging the stream directly into the fuel:* Shooting a high-velocity stream into burning liquid splashes burning fuel across the ground and breaks up the foam blanket!
- ❌ *Partially closing the nozzle bail:* Gating or partially throttling the nozzle increases downstream backpressure, instantly halting foam induction at the eductor.
- ❌ *Kinked hose downstream:* Any kink creates backpressure that causes the eductor to stop picking up concentrate.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Why did the eductor stop pulling foam when the nozzle bail was half closed?", "Which application technique produces the gentlest foam blanket?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Flush the eductor: Submerge pick-up tube in clean bucket of water and flow line for 60 seconds to wash corrosive foam out of the ball check valve.
- ☐ Hose drained, washed, and repacked.
- ☐ Concentrate pails sealed tightly and stowed upright in compartment.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 26

NFPA 1001 Level 2 (JPR 5.3.3) / NFPA 58

ADVANCED FIRE SUPPRESSION

FLAMMABLE GAS / PROPANE TANK COOLING & VALVE SHUTOFF

⏱️ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear, SCBA (On Air) & Flash Hood

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | 2 Handline Attack Teams (4–6 Firefighters)

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to safely manage a pressurized **LPG / Propane Tank** fire or leak, recognize signs of an impending **BLEVE (Boiling Liquid Expanding Vapor Explosion)**, advance toward the tank using **dual interlocking wide-fog streams** for thermal shielding, cool the vapor space of the cylinder, and execute a manual shutoff of the main control valve. Propane tanks (100-lb cylinders, 500-to-1,000 gallon domestic "pigs") are universal heating fuel sources across rural New Brunswick.

📦 TOOLS & APPARATUS SETUP

- Pumper supplying two 1.75" (45mm) attack lines at 100 PSI nozzle pressure.
- 2 combination fog nozzles set to wide-fog shield pattern (minimum 30–60 degrees).
- Training propane tank prop (20-lb BBQ tank, 100-lb cylinder, or scrap 500-gallon tank simulator).
- Water supply established (minimum 1,500+ gallons on site before starting live water evolutions).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **THE BLEVE DANGER:** If flame impinges directly on the **vapor space** (top 20–50% of the tank) without cooling water, the steel weakens and can violently rupture within **8 to 15 minutes**.

⚠️ **BLEVE Warning Signs:** A rising pitch in the sound of roaring gas venting from the relief valve, or visible discoloration/bulging of the tank shell. **EVACUATE IMMEDIATELY (Minimum 500 metres / 1,500 feet)!**

⚠️ **Continuous Water Supply Mandatory:** Never approach a burning pressurized vessel unless an uninterrupted, positive water supply is guaranteed.

🛑 **THE STOP RULE: Shout "STOP!" immediately if either hose line loses water pressure, if nozzles fall out of interlock, or if a firefighter trips during the approach.**

PHASE 1: PROPANE PROPERTIES & APPROACH STRATEGY (15 MINUTES) — SLOW DRY RUN

1. **Properties of Propane (LPG):**
   - Vapor density is **1.5 (heavier than air)**; leaking gas pools in low ditches, basements, and around foundations.
   - Expansion ratio: **270 to 1** (1 liter of liquid expands into 270 liters of explosive gas!).
2. **The Dual-Line Interlocking Shield:**
   - Two 1.75" lines advance side-by-side.
   - Both nozzles set to **wide-fog pattern (45–60 degrees)**.
   - The two fog streams **overlap by 1 to 2 feet in the center**, creating a solid, impenetrable thermal water wall protecting the firefighters behind them.
   - The fog stream pushes radiant heat and flames away while entraining fresh air for the crew.

PHASE 2: COORDINATED APPROACH WALKTHROUGH (15 MINUTES) — BUDDY COACHING

1. Both nozzle teams kneel or crouch side-by-side, shoulder-to-shoulder.
2. Team Leader (between the two nozzlemen) coordinates cadence: *"Step... Step... Step..."*
3. Move forward at a slow, deliberate pace. Nozzlemen keep streams focused on the **upper shell (vapor space)** of the propane tank to keep the steel cold.
4. If the relief valve is burning, **DO NOT extinguish the flame** until the valve can be shut! Extinguishing the flame before shutting the leak creates an invisible cloud of heavier-than-air gas that will ignite into a catastrophic vapor cloud explosion.

PHASE 3: VALVE ISOLATION & SAFE RETREAT (20 MINUTES) — TIMED / STRESS

1. As the crew reaches arm's length from the tank:
   - Both nozzlemen open their fog patterns wide to deflect flames around the sides.
   - The Team Leader (or backup firefighter) reaches between the streams with gloved hands.
   - Grasp the handwheel of the service valve and **rotate clockwise until fully closed**.
2. **Confirm Extinguishment:**
   - Once the valve is closed, the flame will starve and extinguish.
   - Continue cooling the hot tank shell with fog streams until steam generation stops.
3. **The Coordinated Retreat:**
   - **NEVER TURN YOUR BACK ON THE TANK!**
   - Keeping both fog streams flowing toward the tank, the crew backs up step-by-step in unison until clear of the hot zone (minimum 50 feet away).
   - Nozzles are shut down slowly to prevent water hammer.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Turning off nozzles when the fire goes out:* If the valve didn't seal 100%, gas continues leaking; shutting off the water shield leaves you unprotected against re-ignition. Keep the shield up while backing out!
- ❌ *Splitting the fog patterns:* If nozzlemen pull apart, a gap opens in the center of the water shield, allowing superheated flames to blast between them. Keep streams firmly interlocked.
- ❌ *Approaching from the ends of the tank:* If a cylinder ruptures, the convex end caps shoot outward like missile rockets. **Always approach from the sides (perpendicular to the tank ends)!**

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How loud was the water shield roaring during the approach?", "Why must we approach the sides of the tank rather than the ends?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Hoses drained, dried, and repacked.
- ☐ Nozzles flushed and set to department default pattern.
- ☐ SCBA facepieces disinfected; cylinders refilled.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 27

ADVANCED TACTICS

NFPA 1001 Level 2 (JPR 5.3.1, 5.3.2) / UL FSRI Research

COORDINATED FIREGROUND VENTILATION & FLOW PATH CONTROL

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Interior Attack Team & Exterior Vent Team

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must understand modern fire dynamics (UL/NIST research): how opening doors and windows creates a **Flow Path** feeding fresh oxygen to ventilation-controlled fires, how uncontrolled ventilation triggers rapid flashovers, and how to execute **Coordinated Ventilation** synchronized with water application (including exterior transitional attack).

📦 TOOLS & APPARATUS SETUP

- Apparatus pumper with charged 1.75" (45mm) attack line (smooth bore or combination nozzle).
- Ventilation tools: Halligan bar, flathead axe, 6-ft pike pole.
- Station bay doors, window prop, or simulated structure layout.
- Smoke machine or cardboard flow-path demo box (dollhouse prop).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **The Flashover Clock:** In modern homes filled with petroleum-based synthetic furnishings (plastics, poly foam), opening a door or breaking a window to a ventilation-limited fire accelerates flashover in as little as **90 to 120 seconds**.

⚠️ **Wind-Driven Hazards:** Never vent on the upwind side if crews are operating inside! Wind forces superheated gases directly onto interior firefighters.

🛑 **THE STOP RULE: Shout "STOP!" immediately if an exterior window or door is opened before the interior nozzle crew is in position with a charged, flowing line.**

PHASE 1: UNDERSTANDING THE FLOW PATH & MODERN FIRE BEHAVIOR (15 MINUTES) — SLOW DRY RUN

1. **Legacy vs Modern Fires:**
   - Legacy homes (natural wood/cotton): Fire was fuel-limited; took 20+ minutes to flashover.
   - Modern homes (plastics/synthetics): Rapidly consume available oxygen and become **ventilation-limited (oxygen-starved)** while producing massive volumes of unburned fuel gases.
2. **The Flow Path Definition:**
   - The space between an air intake (open door/window) and an exhaust opening (chimney, roof hole, or upper window).
   - **Bidirectional Flow Path:** Fresh oxygen rushes in low; black superheated smoke pushes out high (neutral plane).
   - **Unidirectional Flow Path:** Opening where air is strictly entering (intake) or strictly exhausting under high pressure (blowtorch exhaust).

PHASE 2: DOOR CONTROL & FLOW PATH ISOLATION (15 MINUTES) — BUDDY COACHING

1. **The Entry Door:** The front door is a massive oxygen intake.
2. **Door Control Drill:**
   - Forcible entry team pops the door.
   - **Immediate Action:** Pull the door almost completely shut again, leaving only a 2-inch crack while awaiting the charged hoseline.
   - Chock or hold the door closed to starve the fire of fresh air while the nozzle team flakes hose and dons facepieces.
   - When attack team is on air with line charged: open door, enter low, and control the door behind the hose to minimize incoming air.

PHASE 3: SYNCHRONIZED COORDINATED ATTACK (TRANSITIONAL ATTACK) (20 MINUTES) — TIMED / STRESS

1. **The Exterior "Transitional" Knockdown:**
   - Common in rural NB when arriving with 2–3 firefighters who cannot legally enter IDLH under WorkSafeNB 2-in/2-out rules.
   - Position at an exterior window showing heavy fire.
   - Direct a **solid or straight stream steeply upward against the ceiling** of the fire room.
   - Flow for **15 to 30 seconds** without moving the stream around.
   - Water coats the ceiling, turns to steam, absorbs heat, and dramatically knocks down the fire without pushing heat into adjacent rooms.
2. **Coordinating the Interior Advance with Ventilation:**
   - Exterior vent team stands by at the designated exhaust window with Halligan/pike pole.
   - **DO NOT BREAK THE WINDOW YET.**
   - Interior nozzle team enters, advances to the fire compartment, and begins cooling the ceiling gases with water.
   - Interior Team Leader calls over radio: *"Interior to Vent, we have water on the fire! Take the Side Bravo window!"*
   - Vent team clears the window completely (glass, sash, and curtains) to establish clean exhaust.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Ventilating before water is ready:* Breaking windows while the engine is still laying hose feeds fresh oxygen to the fire, causing rapid structural flashover.
- ❌ *Failing to clear the window sash:* Leaving window sashes or heavy blinds in place reduces ventilation efficiency by over 50%. Clean the entire window opening from top to bottom!
- ❌ *Whipping the nozzle in circles during transitional attack:* Rapidly swirling the nozzle creates high air entrainment, acting like a giant fan that pushes heat throughout the house. Hold the stream still against the ceiling!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"What happens to an oxygen-starved fire when we open the front door?", "Why must the vent team wait for the radio call from the nozzle team before breaking glass?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Pike poles and Halligan bars inspected and remounted.
- ☐ Broken glass or prop debris swept and disposed of.
- ☐ Hose drained, rolled, and repacked.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS

DRILL 28

NFPA 1001 Level 2 (JPR 5.5.1) / NFPA 1962

APPARATUS & EQUIPMENT MAINTENANCE

ANNUAL SERVICE HOSE TESTING & INSPECTION

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Pumper / Hose Testing Rig | 🦺 **PPE:** Helmet, Safety Glasses, Boots & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Testing & Inspection Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to inspect fire hose for cuts, chemical degradation, coupling damage, and missing gaskets, connect hose to a test manifold or pumper discharge with a safety test gate valve, bleed all air from the lines, pressurize the layout to **NFPA 1962 service test pressure**, hold the pressure for **5 minutes**, and safely mark or tag out defective sections. A burst hose line inside a burning building is catastrophic.

📦 TOOLS & APPARATUS SETUP

- Pumper or specialized electric hose tester with test gate valve (valve with 1/4" bleeder hole to prevent pressure surges if a hose bursts).
- Sections of 1.75" (45mm) attack line and 4"/5" (100mm/125mm) LDH supply line.
- Test nozzles or threaded test caps with petcock bleeder valves.
- Spanner wrenches, grease pencil/paint marker, hose record log sheets.
- 100-ft tape measure.

⚠ SAFETY WATCH-OUTS & STOP RULE

⚠ **Catastrophic Burst Hazard:** During the 5-minute pressurized hold, **NO PERSONNEL MAY STRADDLE OR WALK ALONG THE HOSE LINES!** Stand a minimum of **15 feet to the side** (the "safe zone").

⚠ **Trapped Air Hazard:** Air compresses under pressure; water does not. Trapped air turns a bursting hose into an explosive whip. **ALL AIR MUST BE COMPLETELY BLED** before raising pressure!

🛑 **THE STOP RULE: Shout "STOP!" immediately if anyone approaches a pressurized line, or if a coupling begins to slip noticeably from the hose collar.**

PHASE 1: VISUAL INSPECTION & COUPLING CHECK (15 MINUTES) — SLOW DRY RUN

1. Lay hose straight on flat pavement.
2. **Jacket Inspection:** Check for mold/mildew rot, cuts, heavy abrasion, burns, or chemical stains (battery acid, motor oil).
3. **Coupling & Gasket Inspection:**
   - Swivel must rotate smoothly by hand without binding.
   - Inspect female swivel gasket: Must be supple, free of cracks, and seated flat in the groove.
   - Check lugs and threads for dents, burrs, or flattening.
   - Check Storz locks for spring tension and intact locking levers.
4. **Marking the Coupling Seam:** Draw a line with a grease pencil around the hose jacket directly where it meets the metal coupling ferrule. (If the coupling slips during pressure testing, this pencil line will move, revealing a failing connection!).

DRILL 28: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Apparatus & Equipment Maintenance • Practical Application

PHASE 2: LAYING OUT & BLEEDING AIR (20 MINUTES) — BUDDY COACHING

1. **Layout Rules:**
   - Maximum length of any single test line connected to a discharge is **300 feet (90m)**.
   - Keep lines straight without sharp kinks or bends.
2. **Connecting & Bleeding:**
   - Connect hose to pumper test gate valve or hose tester manifold.
   - Connect test cap or shutoff nozzle to the far end.
   - Open the test gate valve and charge the layout with low hydrant/tank pressure (~45–50 PSI).
   - Elevate the far end test cap/nozzle slightly and open the bleeder petcock until all trapped air escapes in a steady stream of solid water.
   - Close the bleeder valve and verify the test line is completely full of water with zero air pockets.

PHASE 3: PRESSURIZATION, 5-MINUTE HOLD & LOG SIGN-OFF (20 MINUTES) — TIMED / STRESS

1. **Service Test Pressures (NFPA 1962 Standard):**
   - **Attack Hose (1.75" / 2.5"):** Typically tested at **250 to 300 PSI** (or original stenciled service test pressure).
   - **Supply Line / LDH (4" / 5"):** Typically tested at **200 PSI**.
2. **Ramping Pressure:**
   - Raise pump pressure slowly at a rate not exceeding **25 PSI per second**.
   - Once target service test pressure is reached, close the test gate valve.
   - Start the timer: Hold pressure for **5 MINUTES**.
   - Personnel monitor lines from the designated safe zone 15 feet away.
3. **Post-Test Inspection:**
   - After 5 minutes, bleed pressure down slowly to zero.
   - Inspect coupling pencil marks: If the line moved more than 1/8 inch (3mm) away from the coupling, or if water weeping occurred around the shank, the hose **FAILS**.
   - If any leaks, pinholes, or coupling slippage occur: Cut the defective hose coupling off or spray-paint a red "X" and tag out of service.
   - Record hose ID numbers, test date, test pressure, and pass/fail status in the department NFPA log.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Testing hose with trapped air:* Air acts as a loaded spring. A burst hose with trapped air will explode and throw shrapnel; a hose with 100% water simply splits with a harmless splash.
- ❌ *Walking over charged test lines:* High-pressure testing exposes latent weaknesses in fabric. Never step over or straddle lines under test!
- ❌ *Ramping pressure too fast:* Spiking pressure instantly damages internal rubber liners and blows out pump packing.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did any pencil marks move on our older hose couplings?", "Why must hose be tested annually under NFPA standards?"***

✅ APPARATUS RESET CHECKLIST

- [ ] All passed hose sections stamped/logged with test year and stowed back on apparatus beds.
- [ ] Any failed hose tagged, marked with red tape, and placed in repair bin.
- [ ] Test caps and manifold dried and returned to maintenance cabinet.
- [ ] Hose test log signed and filed for Fire Marshal / FUS records.
- [ ] Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS